MEssentials: A Centralized Student Hub for UP Diliman's Department of Mechanical Engineering

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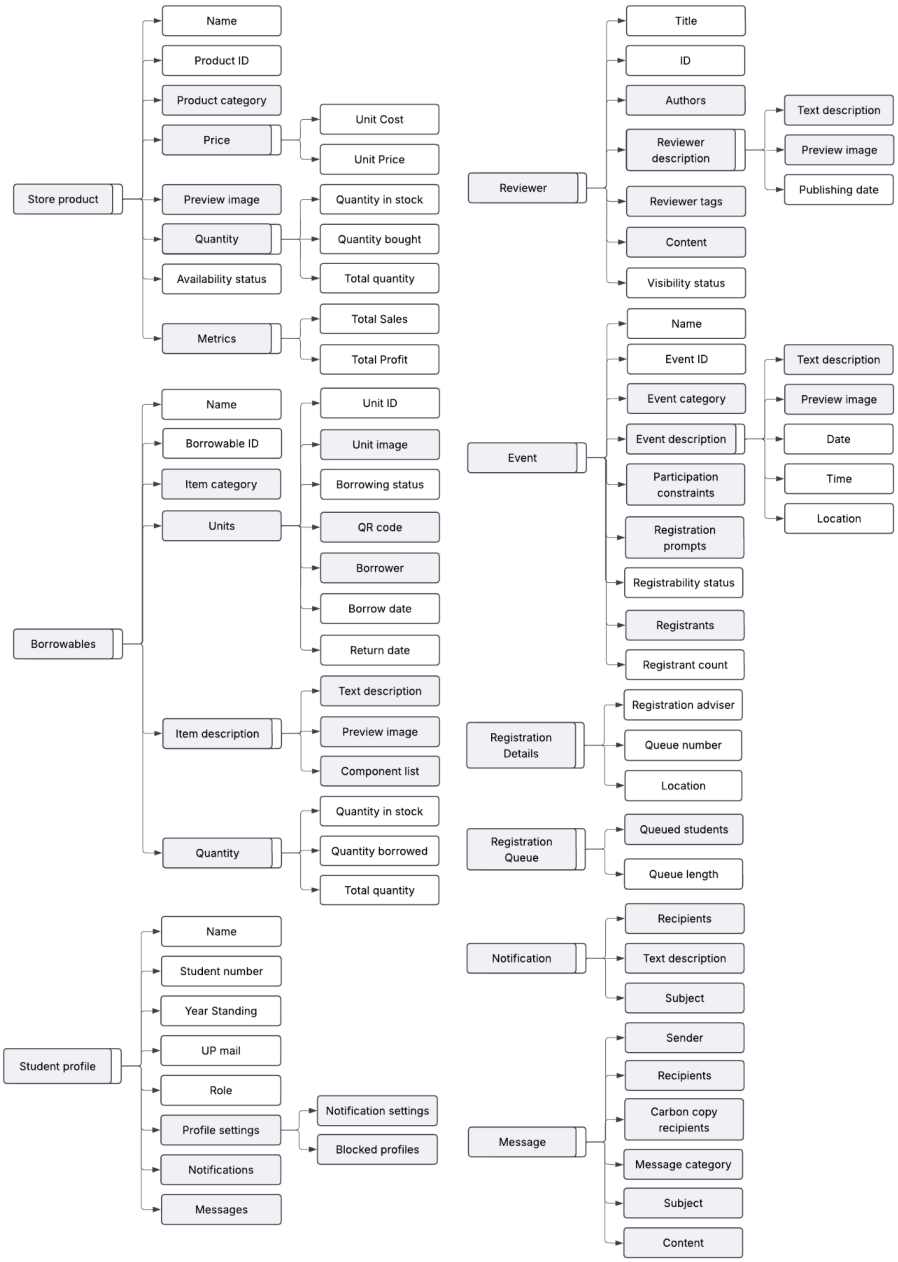
Submitted to:

Dr. Ligaya Leah L. Figueroa

Web App Requirements Model

Content Model

This is the collection of data trees detailing the hierarchy, attributes, and relationships among content objects utilized by MEssentials:



Interaction Model

Actors

There are three primary actors involved in the system:

- **Students** are *participating* actors who aim to utilize the student services hosted on the website, including the HonestME Store, ME-kailangan, MEg-aral, ME Days, and registration queueing.
- **Volunteers** are *initiating* actors who aim to facilitate said student services hosted on the website. They can further be split into specific roles, each managing a specific student service:
    - HonestME Store Managers
    - ME-kailangan Managers
    - MEg-aral Managers
    - ME Days Managers
    - Registration Managers
- **Administrators** are *initiating* actors who aim to control the roles and privileges of the volunteers.

Use-cases

The **students** participate in the following use cases:

- Profile use cases:
    - *LogIn* - Log in to their student profile through their UP mail.
    - *UpdateProfile* - Update the details in their student profile.
- Notification use cases:

    - *ToggleNotifs* - Toggle notifications for specific student services (ie. HonestME Store restocks).
    - *ViewNotifs* - View notifications sent to you by the website.
    - *DeleteNotif* - Delete notification(s) from your notification list.
- Ticket use-cases:
    - *ViewTickets* - View tickets that you've sent or are involved in.
    - *DeleteTicket* - Delete ticket(s) from your notification list.
    - *SendVolunteerTicket* - Send messages (ie., questions, concerns, feedback) about a specific student service, the website, or other matters to their respective managers, all volunteers, and/or administrators.
- HonestME Store use cases:
    - *ViewStore* - View all products in the HonestME Store, either on a personal device or on the designated store device interface.
    - *SearchStore* - Search for a specific product by its name using the search bar.
    - *FilterStore* - Filter the products based on category or availability.
    - *CartProduct* - Add a product to the cart to be purchased later on the store device interface.
    - *UncartProduct* - Remove a product from the cart on the store device interface.

    - *CheckOutCart* - Check out all products in the cart on the store device interface.
- ME-kailangan use cases:
    - *ViewBorrowables* - View all items for borrowing in ME-kailangan, either on a personal device or on the designated borrowing station interface.
    - *SearchBorrowables* - Search for a specific borrowable item by its name using the search bar.
    - *FilterBorrowables* - Filter the borrowable items based on category or availability.
    - *BorrowBorrowable* - Mark a specific unit of an item as borrowed by you through the borrowing station interface.
    - *ReturnBorrowable* - Mark a specific unit of an item borrowed by you as returned through the borrowing station interface.
- MEg-aral use cases:
    - *ViewReviewers* - View all visible reviewers in MEg-aral.
    - *SearchReviewers* - Search for a specific reviewer by name using the search bar.
    - *FilterReviewers* - Filter the reviewers based on tags.
    - *OpenReviewer* - View the content of a reviewer through the website.
    - *ToggleReviewerBookmark* - Enable or disable a reviewer bookmark, used for quick access.

    - *DownloadReviewer* - Download the content of a reviewer for offline viewing.
- ME Days use cases:
    - *ViewEvents* - View all events under ME Days.
    - *SearchEvents* - Search for a specific event by its name using the search bar.
    - *FilterEvents* - Filter the events based on categories or registrability.
    - *ToggleEventRegistration* - Register for or unregister from a future event.
- Registration use cases:
    - *QueueForRegistration* - Enter the online queue for registration during the registration period and obtain a queue number.
    - *ViewRegistrationDetails* - View registration details, including your place in the queue and registration adviser.

Meanwhile, each **volunteer role** has the following additional privileges:

- All roles:
    - *SendStudentTicket* - Send a response (ie., reminder, reply) to a student profile, with other volunteers and administrators as carbon copy recipients.
- HonestME Store Managers:
    - *AddProduct* - Add a new product in the HonestME Store.
    - *ModifyProduct* - Edit specific attributes of a product.

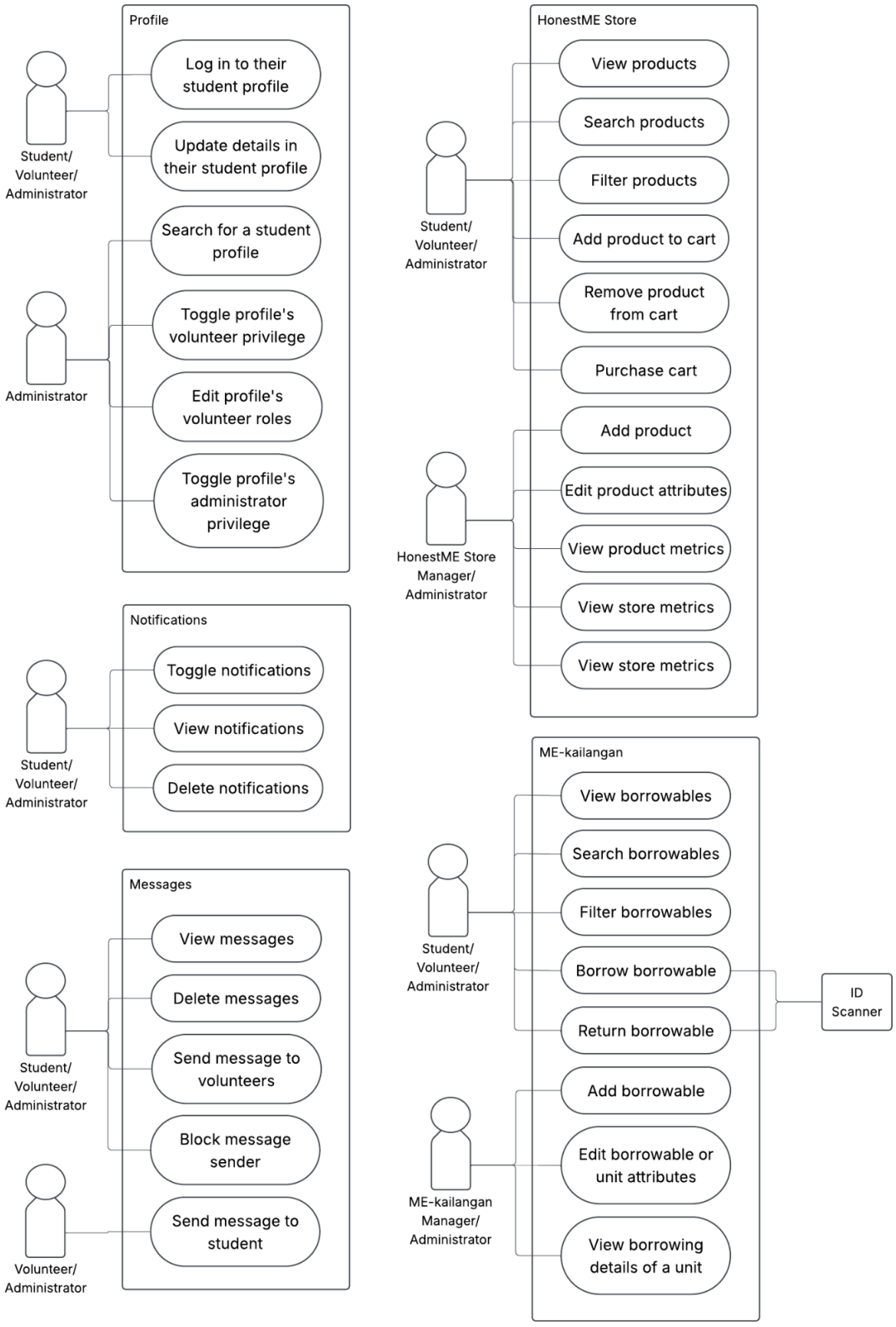
    - *ViewProductMetrics* - View the metrics of a product.
    - *ViewStoreMetrics* - View the metrics of the whole store based on per-product metrics.
    - *SendRestockNotif* - Notify all students of a recent restock.
- ME-kailangan Managers:
    - *AddBorrowable* - Add a new borrowable under ME-kailangan.
    - *ModifyBorrowable* - Edit specific attributes of a borrowable item or one of its specific units.
    - *ViewBorrowingDetails* - View the borrowing details of a specific unit of a borrowable, including the borrower's profile, borrow date, and expected return date.
- MEg-aral Managers:
    - *AddReviewer* - Add a new reviewer under MEg-aral.
    - *ModifyReviewer* - Edit specific attributes of a reviewer.
    - *ToggleReviewerVisibility* - Toggle the visibility of the reviewer among students.
- ME Days Managers:
    - *AddEvent* - Add a new event under ME Days.
    - *ModifyEvent* - Edit specific attributes of an event.
    - *ViewRegistrants* - View the number of registrants for an event alongside their profiles.
- Registration Managers:

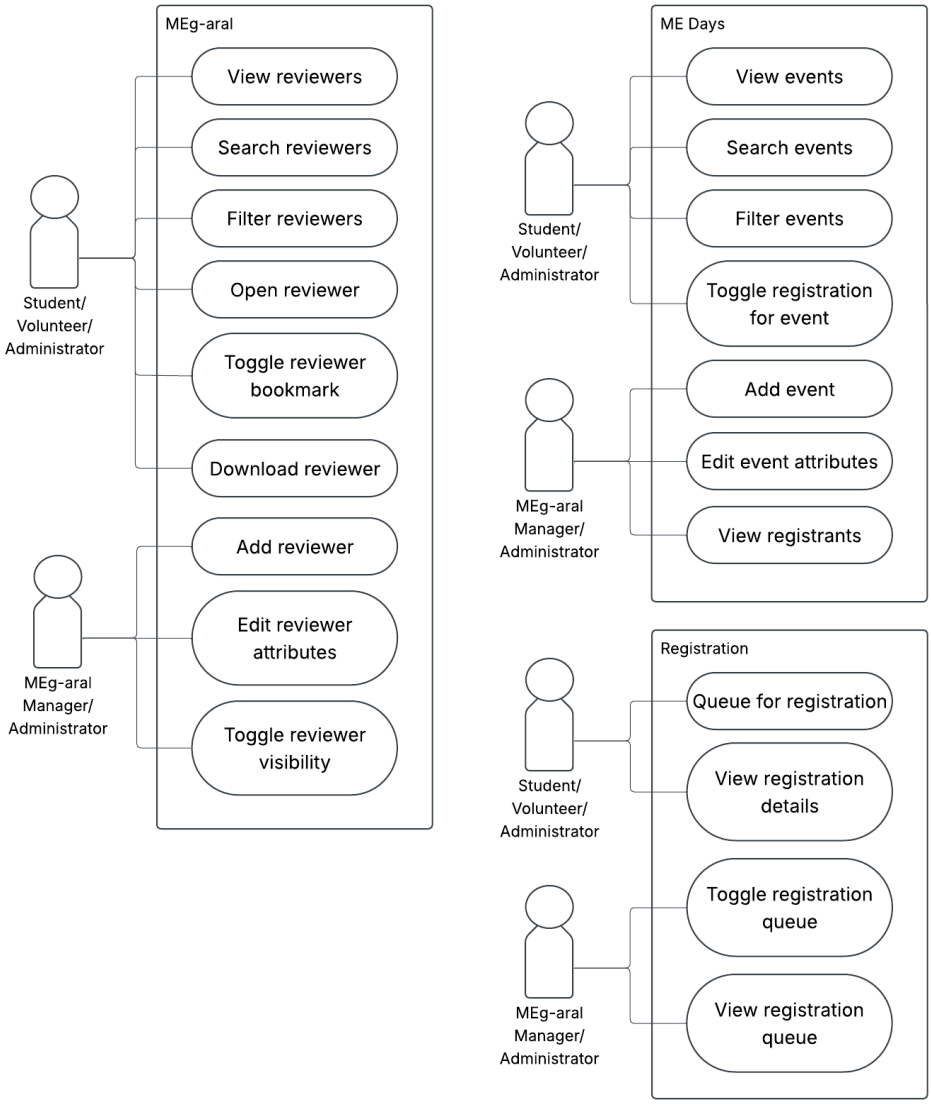
    - *ToggleRegistrationQueue* - Open or close the registration queue for all students.
    - *ViewRegistrationQueue* - View the profiles and registration details of all students in the queue.

Lastly, the **administrators** have the privilege of participating in all mentioned use cases, including the following:

- *ToggleVolunteerPrivileges* - Toggle a student profile's volunteer privileges (and roles, if applicable).
- *ToggleAdministratorPrivileges* - Toggle a student profile's administrator privileges, excluding your own.
- *EditVolunteerRole* - Edit the role(s) of a volunteer, in turn editing their privileges.

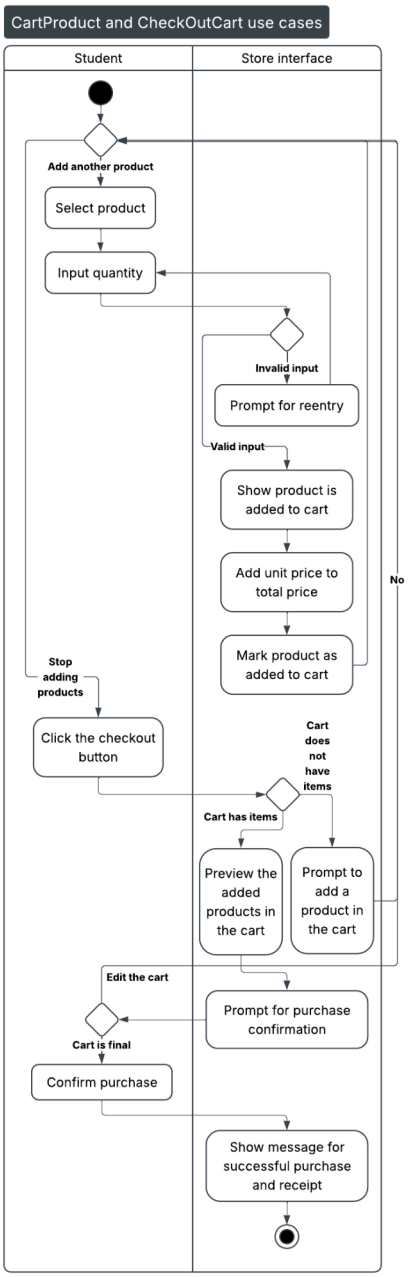
To put it all together, here are the use case diagrams depicting these scenarios:

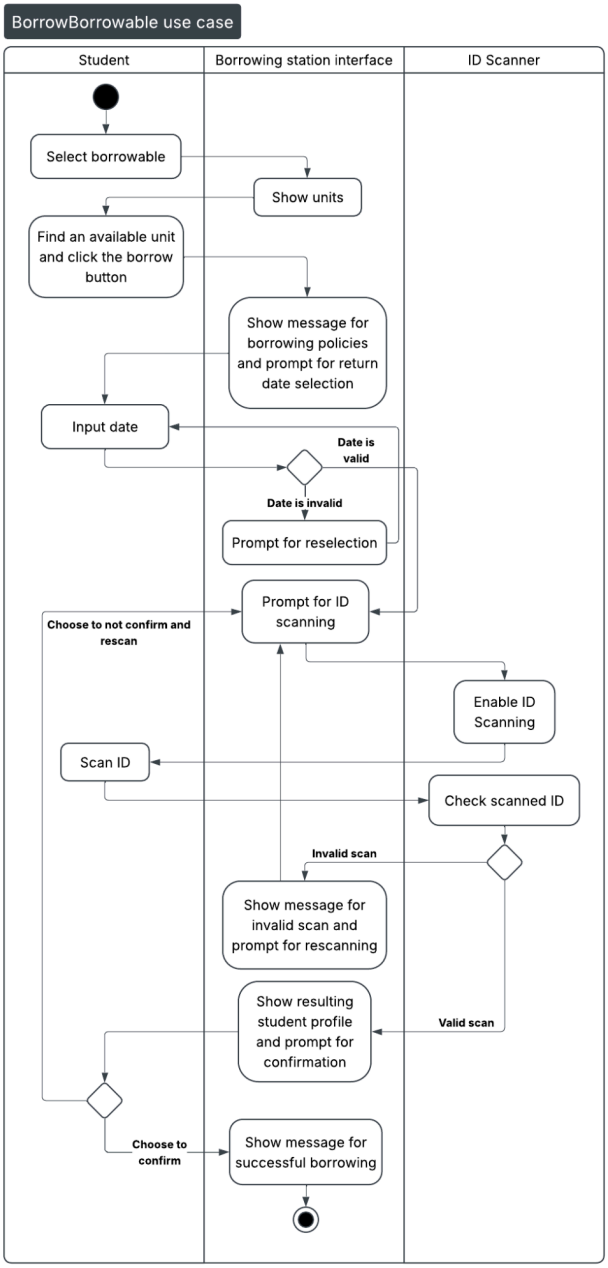



MEg-aral
View reviewers
Search reviewers
Filter reviewers
Open reviewer
Toggle reviewer bookmark
Download reviewer
Add reviewer
Edit reviewer attributes
Toggle reviewer visibility
Student/ Volunteer/ Administrator
MEg-aral Manager/ Administrator
ME Days
View events
Search events
Filter events
Toggle registration for event
Add event
Edit event attributes
View registrants
Student/ Volunteer/ Administrator
MEg-aral Manager/ Administrator
Registration
Queue for registration
View registration details
Toggle registration queue
View registration queue
Student/ Volunteer/ Administrator
MEg-aral Manager/ Administrator

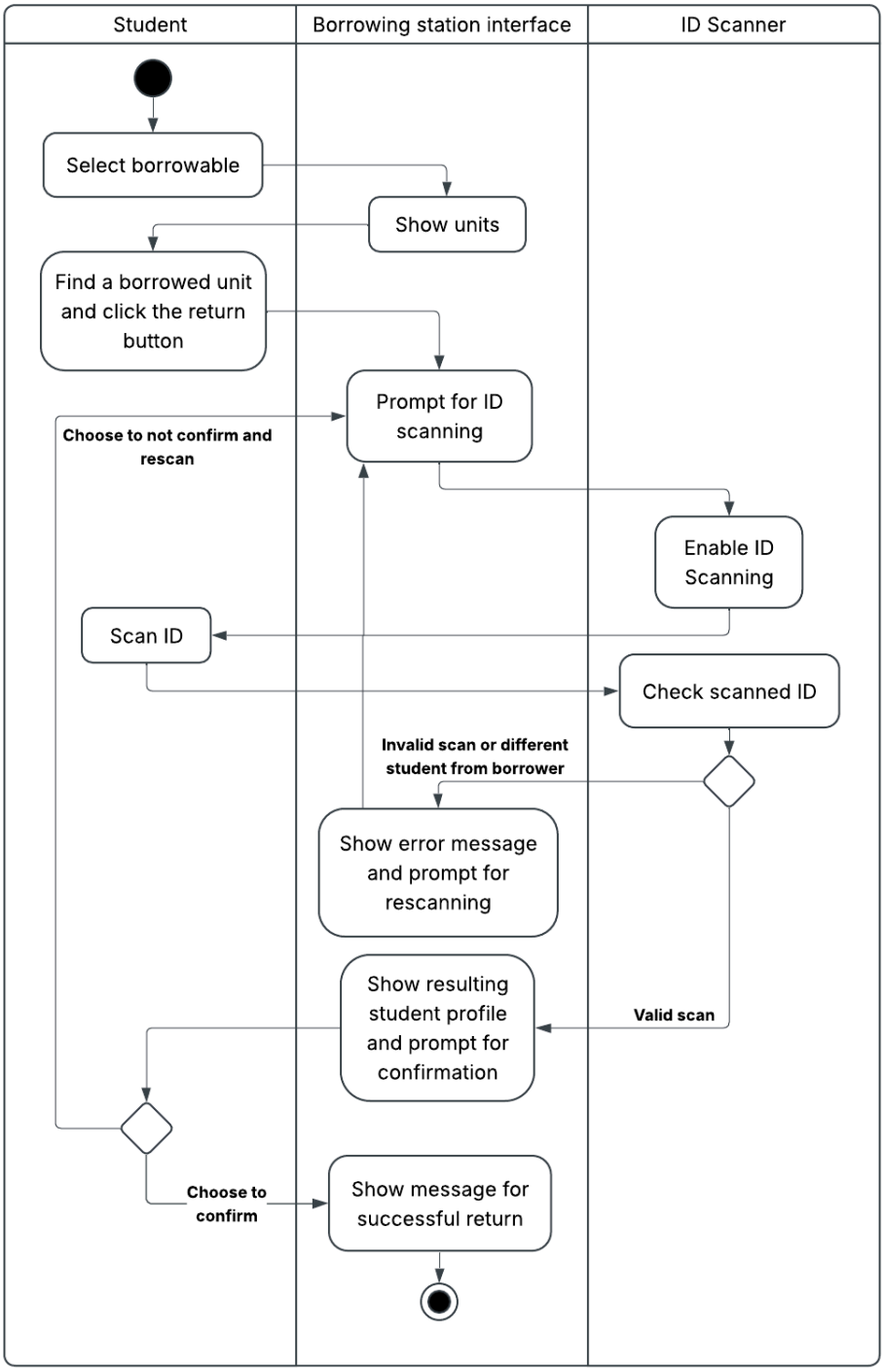
System Sequence Diagrams

To further expound on the important test cases, here are swimlane diagrams for use cases CartProduct, CheckOutCart, BorrowBorrowable, ReturnBorrowable, OpenReviewer, AddReviewer, ToggleEventRegistration, QueueForRegistration, ViewRegistrationDetails, and ToggleVolunteerPrivileges.

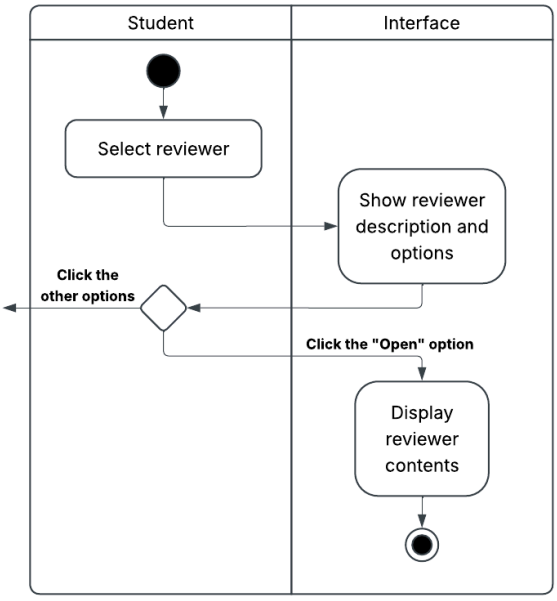




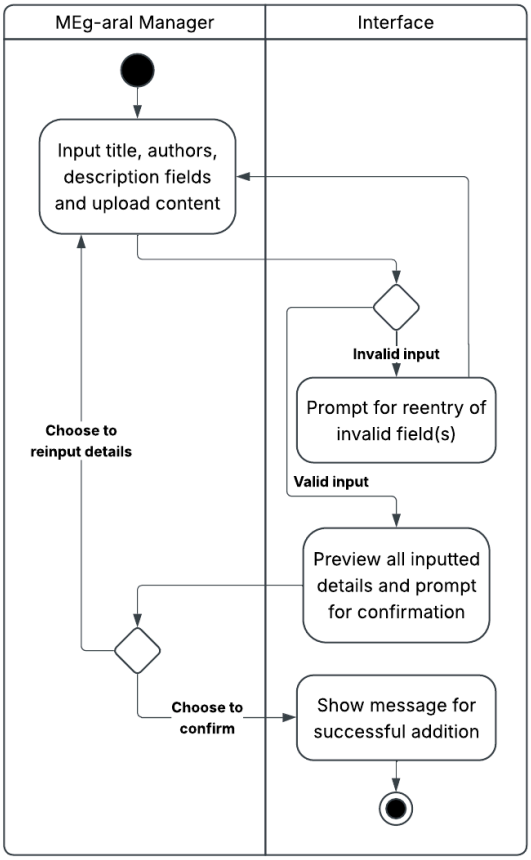
ReturnBorrowable use case



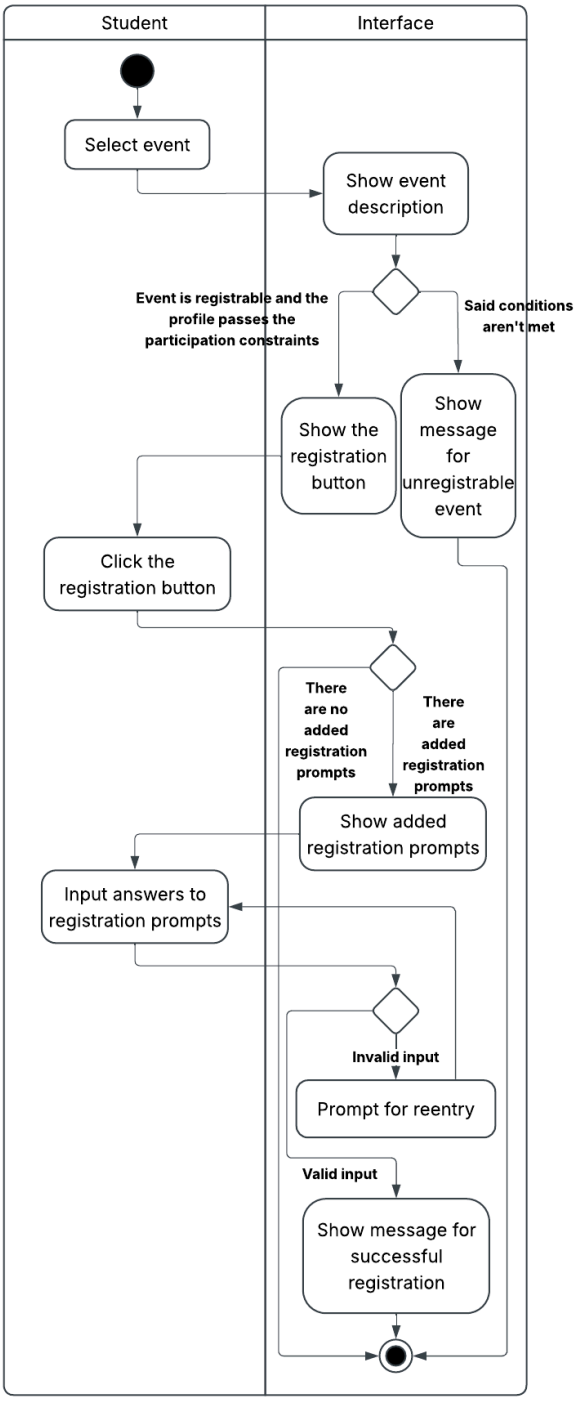
OpenReviewer use case



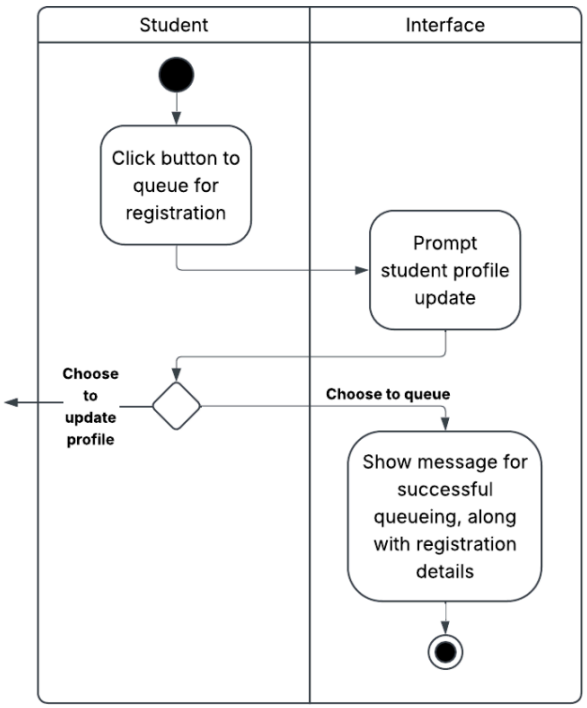
AddReviewer use case



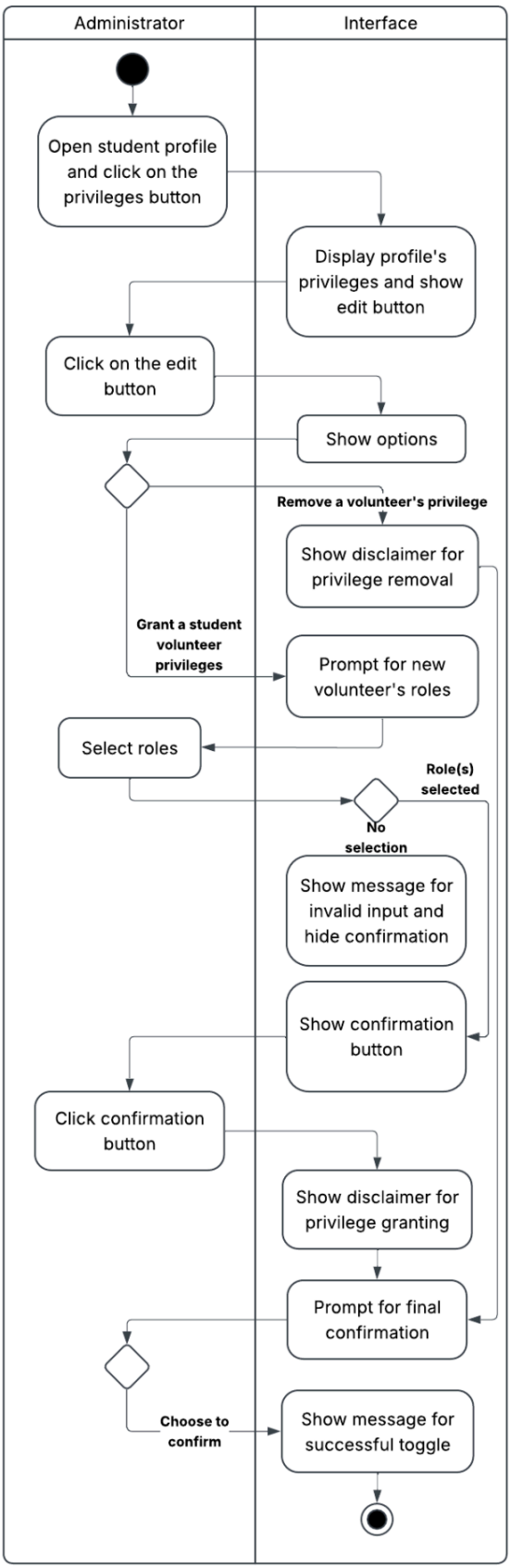
ToggleEventRegistration use case



QueueForRegistration and DisplayRegistrationDetails use case

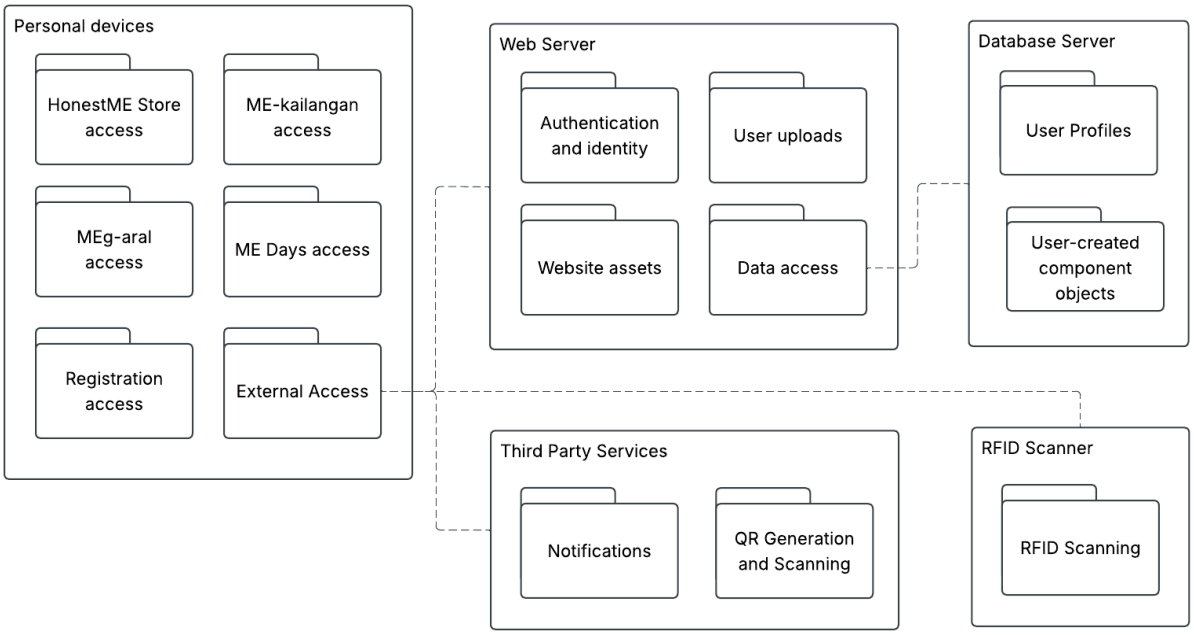


ToggleVolunteerPrivileges use case



Configuration Model

For the configuration model, refer to the following UML deployment diagram, detailing the allocation of software functionality and subsystems within the physical computing environment:



Web App Architecture and Design

Architecture Design

Summary and Status

ID	Decision	Status	Last Update
AD-001	Use a composite structure based on linear and networked structures for the student views	DM	18Oct26
AD-002	Use MVC for the main architecture.	AD	18Oct26

Architectural Decisions

Subject Area	Content Architecture	**Topic**	Content Structures
Architectural Decision	Use a composite structure based on linear and networked structures for the student views	**ID**	AD-001
Issue or Problem Statement	All information presented to users needs to be easy to navigate towards while differentiating the different services well.		
Assumptions	• Most information of the services from the user side have a predictable sequence of interactions that is known. • There are some pieces of information that users will want to frequently navigate between.		

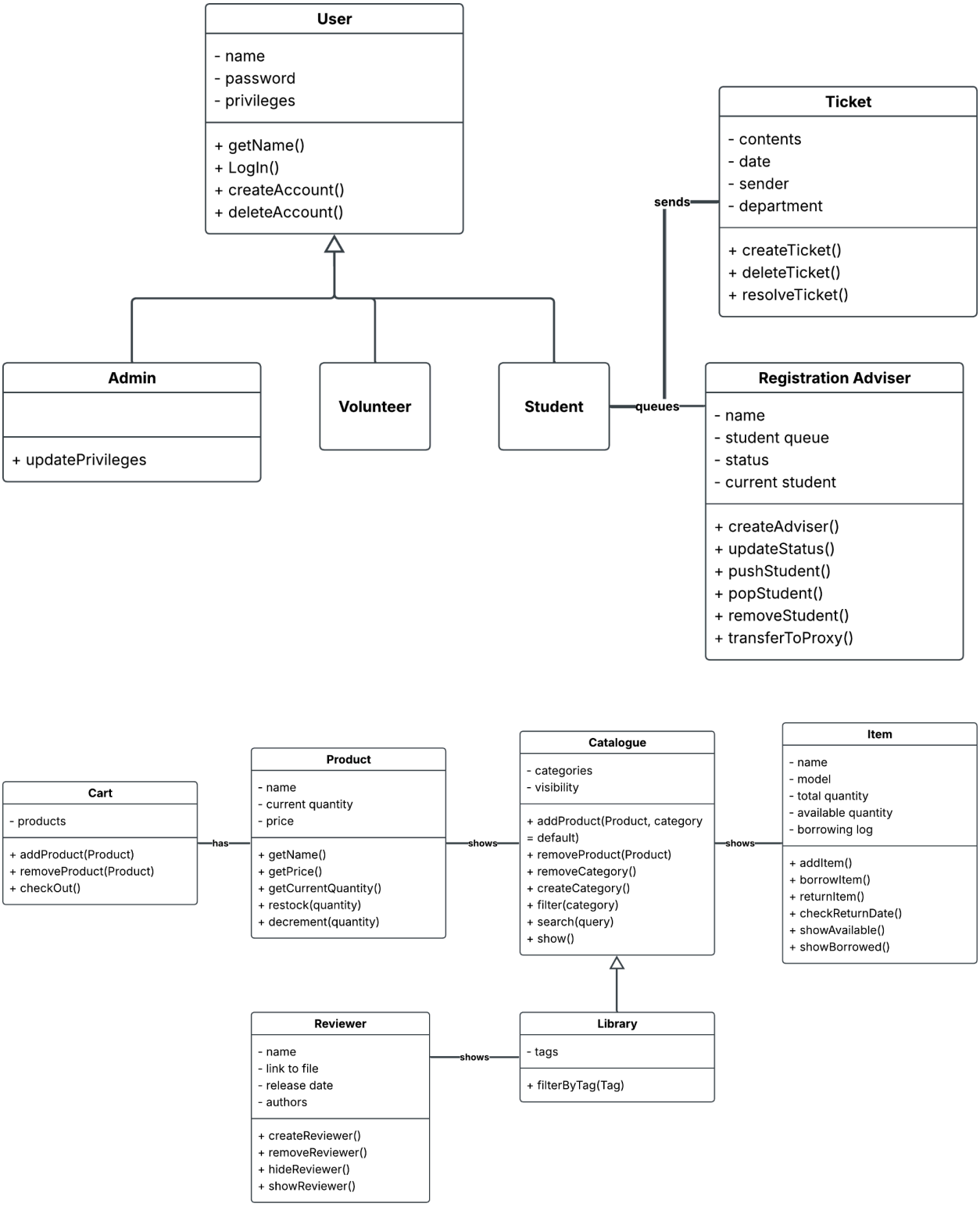
	• Users will want to be able to use a different service easily from any given service.
Motivation	Make site information accessible and achieve all the functionality goals of the quality requirement tree while still maintaining the usability goals.
Alternatives	• Option 1 - go purely linear structure since there is a predictable sequence of interactions. This would be good for making the website understandable since all information is organized well. • Option 2 - use a networked structure. This will allow users to easily access any part of the website they need from anywhere and would make accessing different areas frictionless. • Option 3 - use a linear structure to flow vertically down, but implement a sort of “networked” option to navigate to higher areas from lower areas easily. This would still retain the intuitiveness of a linear structure and allow for users to navigate between services easily.
Decision	Go for option 3.
Justification	It is the option that would allow for the best balance of the quality requirement tree.
Implications	The amount of “higher” areas that can be accessed beads to be decided well so that it’s not overwhelming to navigate.

Subject Area	WebApp Architecture	**Topic**	WebApp Architecture Types
Architectural Decision	Use MVC for the main architecture.	**ID**	AD-002
Issue or Problem Statement	The app needs to be able to be fast and update the database frequently from the borrowing and honesty store stations, while giving a relatively good information estimate to remote users possibly in their own homes. The reviewer information updates need to reflect relatively quickly and also need to be accessed remotely.		

Assumptions	• The organization is not expecting quantities listed remotely to be exactly accurate for the honesty story, but the borrowing needs to be accurate. • Most of the highly updateable systems will be closely monitored by volunteers. • The architecture needs to be relatively scalable.
Motivation	Make sure that the code is maintainable but also is fast enough for the remote access needed.
Alternatives	• Option 1 - use the book's suggested MVC. This is familiar to the developers and may help with code maintenance. • Option 2 - single page application architecture. This is raised for its speed and would basically allow users to use everything without changing pages. • Option 3 - microservice architecture. This would help with the maintenance of the system, especially if different committees of the org are responsible for different areas.
Decision	Go for Option 1.
Justification	This is the most familiar and will help with making development easy.
Implications	MVC isn't exactly the most performant architecture so there may be some issues there.

Content (Data) Design

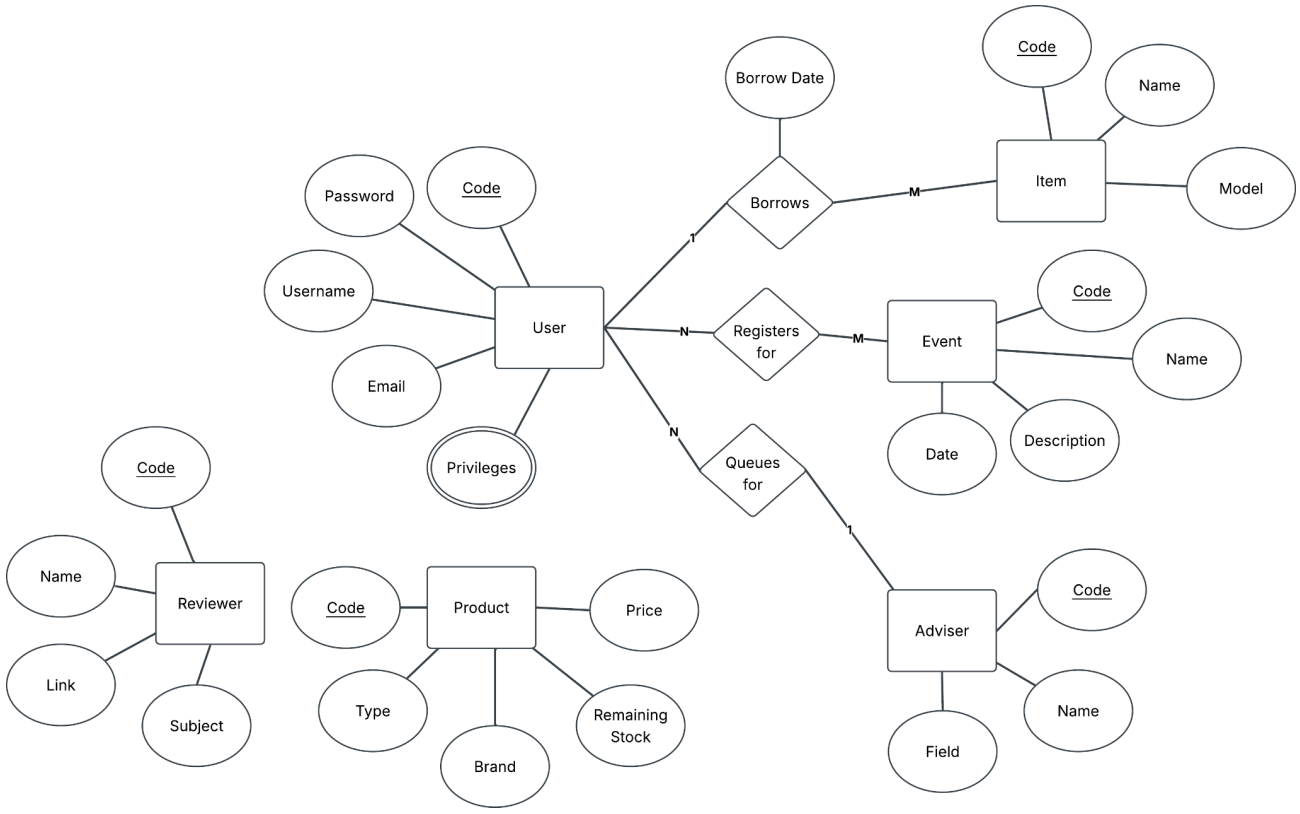
Class Diagram



Event
- date and time - registered users - description
+ createEvent() + removeEvent() + show() + registerUse()

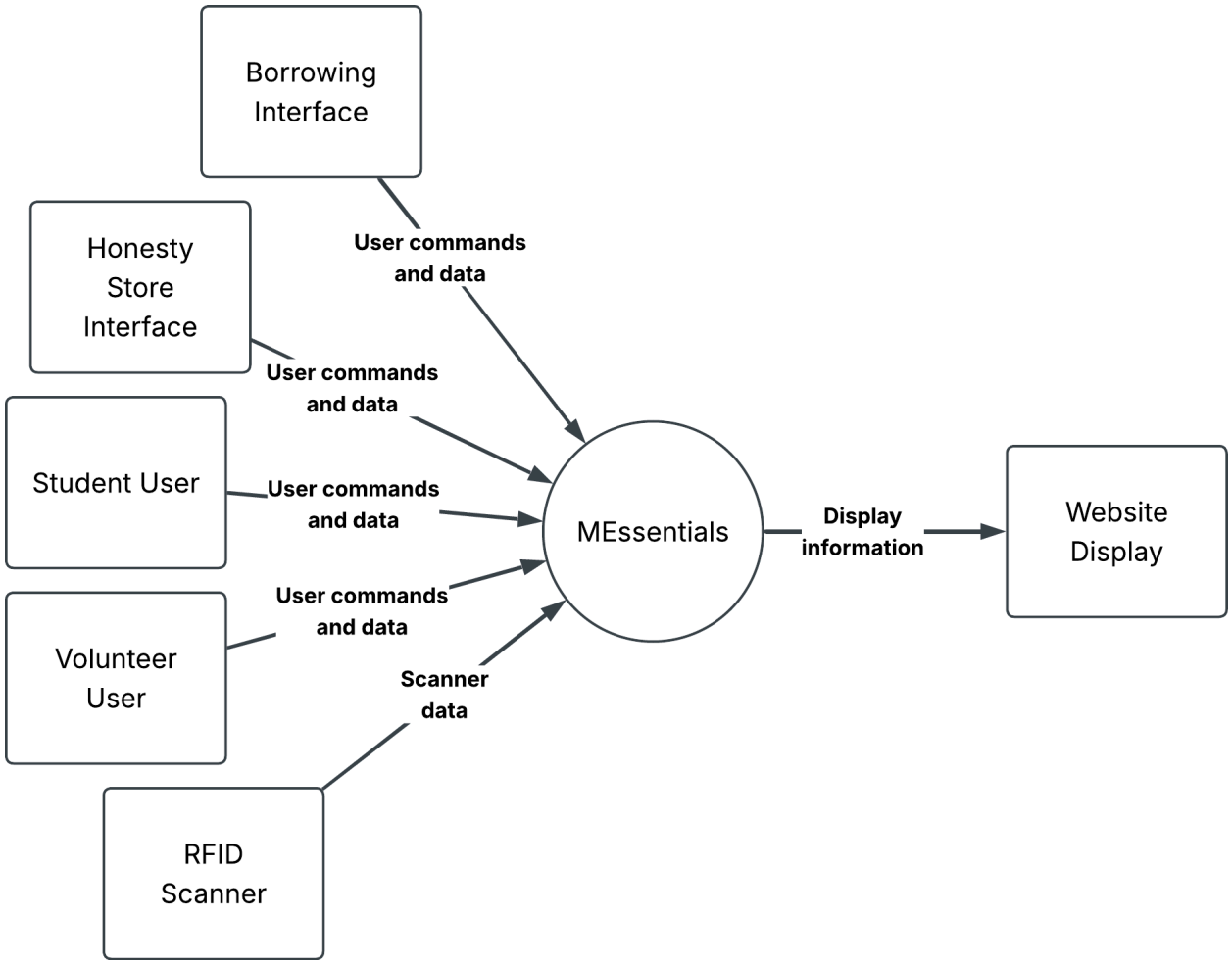
Notification
- name - details
+ init() + toggle() + getDetails() + modify()

Entity-Relationship Diagram

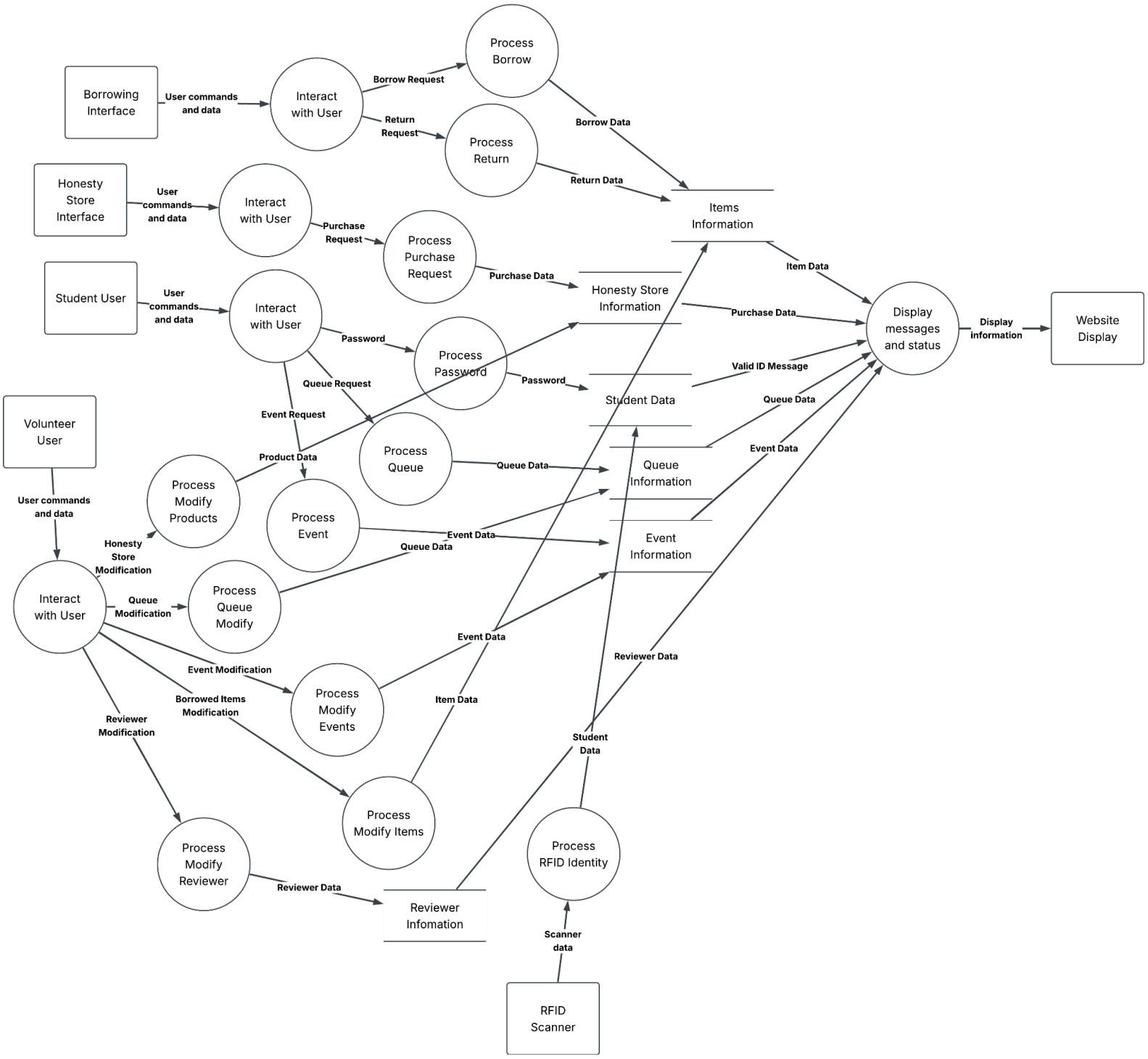


Data Flow Diagram

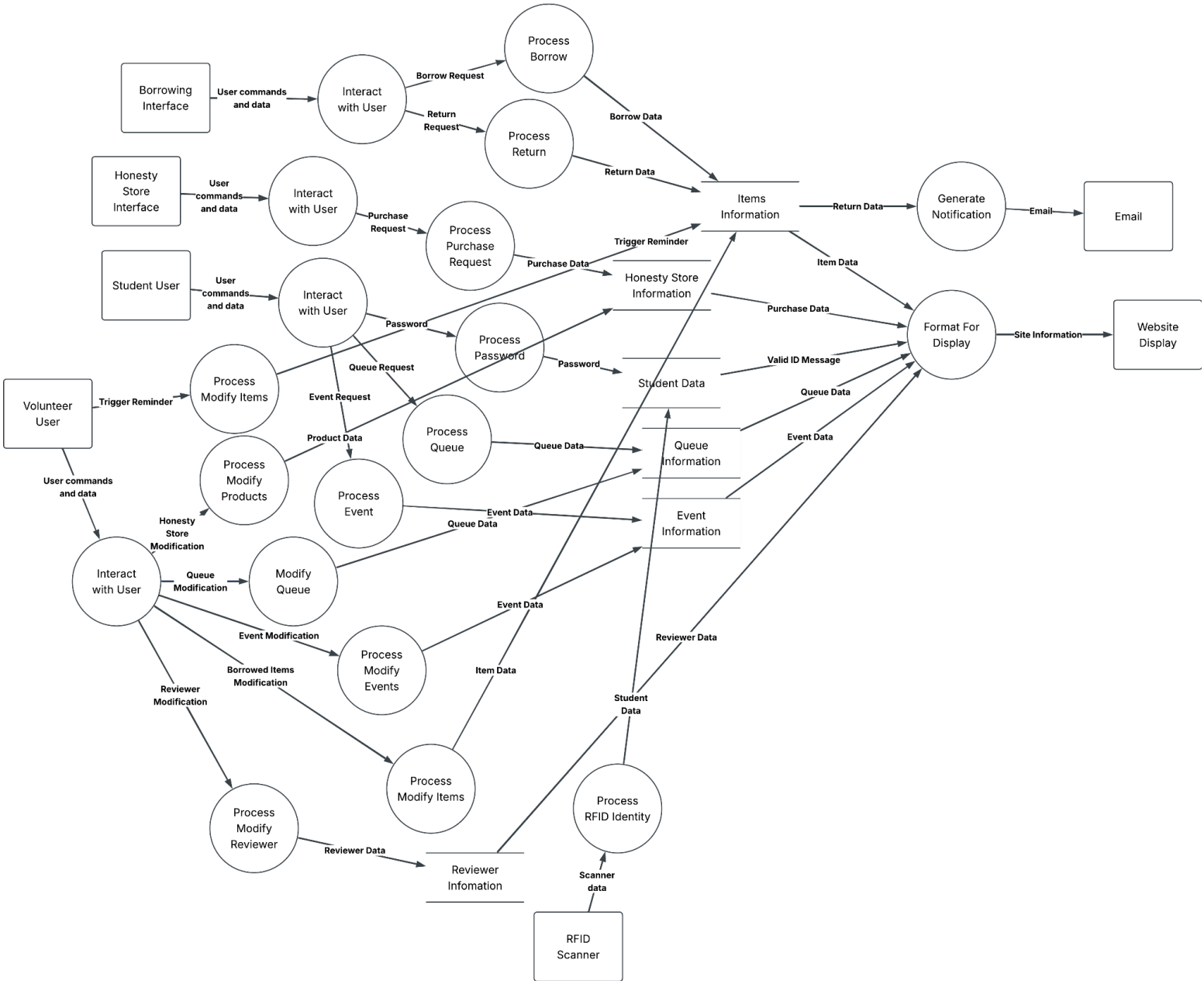
A. Level 0



B. Level 1

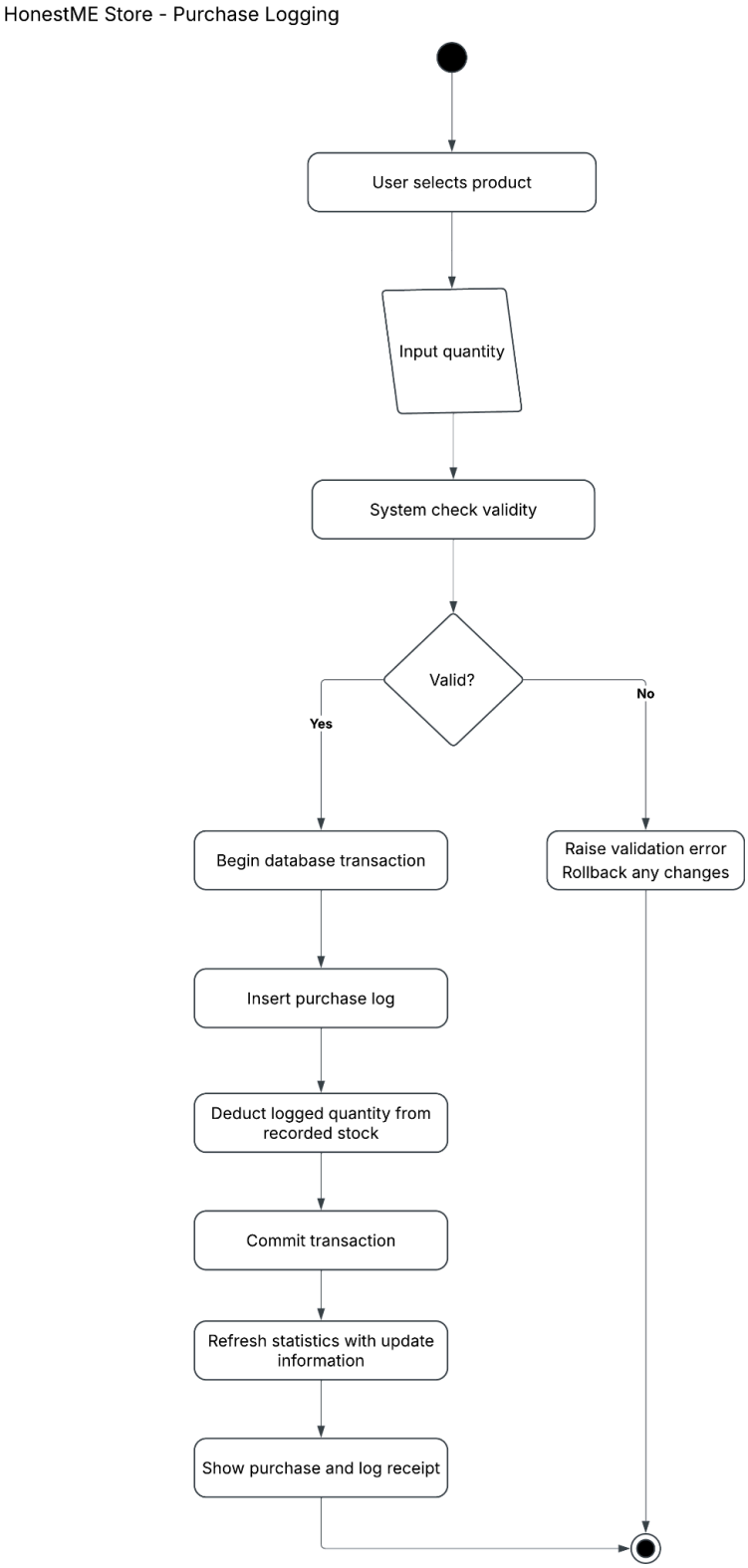

Borrowing Interface
User commands and data
Interact with User
Borrow Request
Return Request
Process Borrow
Process Return
Borrow Data
Return Data
Items Information
Honesty Store Interface
User commands and data
Interact with User
Purchase Request
Process Purchase Request
Purchase Data
Honesty Store Information
Item Data
Purchase Data
Student User
User commands and data
Interact with User
Password
Process Password
Password
Student Data
Valid ID Message
Queue Request
Event Request
Queue Data
Display messages and status
Display information
Website Display
Volunteer User
User commands and data
Product Data
Process Modify Products
Process Queue
Queue Data
Queue Information
Event Data
Process Event
Event Data
Event Information
Queue Data
Honesty Store Modification
Interact with User
Queue Modification
Process Queue Modify
Event Data
Reviewer Data
Event Modification
Borrowed Items Modification
Process Modify Events
Item Data
Reviewer Modification
Student Data
Process Modify Items
Process Modify Reviewer
Reviewer Data
Reviewer Infomation
Process RFID Identity
Scanner data
RFID Scanner

C. Level 2

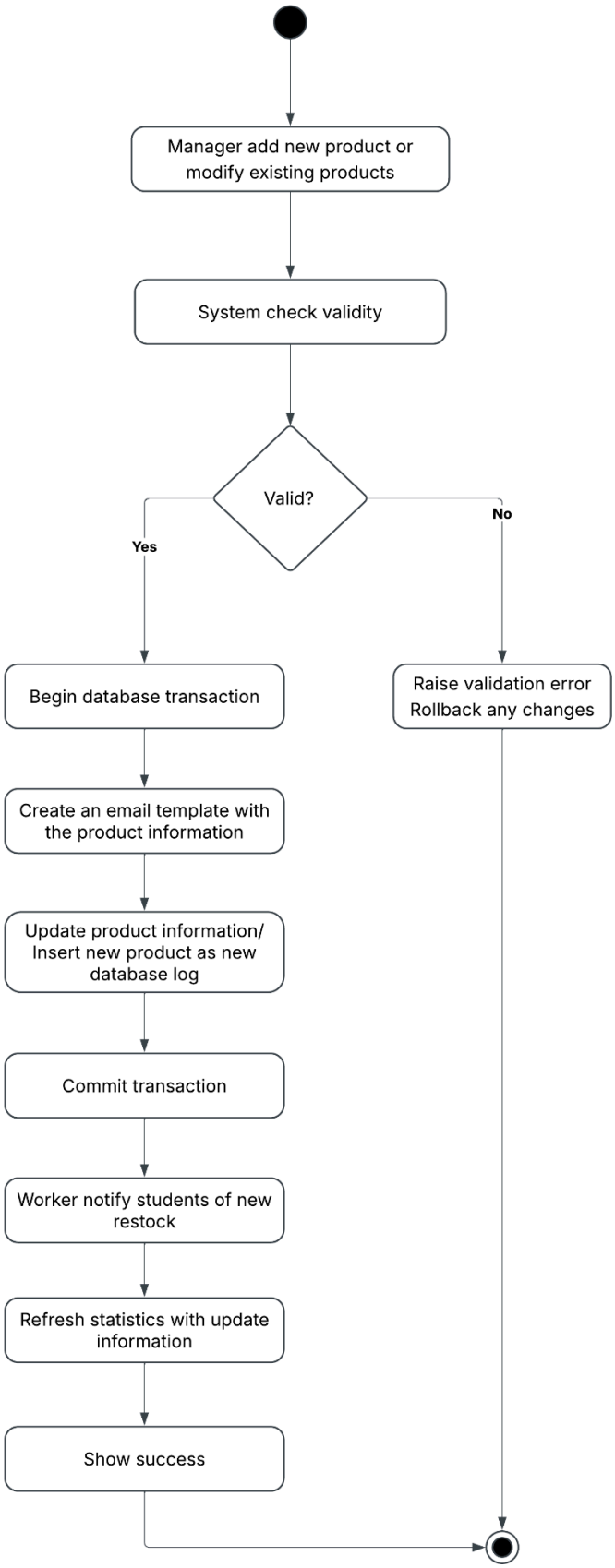

Borrowing Interface
User commands and data
Interact with User
Borrow Request
Process Borrow
Return Request
Process Return
Borrow Data
Return Data
Items Information
Return Data
Generate Notification
Email
Email
Honesty Store Interface
User commands and data
Interact with User
Purchase Request
Process Purchase Request
Purchase Data
Trigger Reminder
Honesty Store Information
Item Data
Student User
User commands and data
Interact with User
Password
Process Password
Password
Student Data
Purchase Data
Format For Display
Site Information
Website Display
Valid ID Message
Volunteer User
Trigger Reminder
Process Modify Items
Event Request
Queue Request
Process Queue
Queue Data
Queue Information
Queue Data
Event Data
Product Data
Process Modify Products
Process Event
Event Data
Event Information
Queue Data
User commands and data
Honesty Store Modification
Interact with User
Queue Modification
Modify Queue
Event Modification
Process Modify Events
Event Data
Item Data
Reviewer Data
Borrowed Items Modification
Reviewer Modification
Student Data
Process Modify Items
Process RFID Identity
Process Modify Reviewer
Reviewer Data
Reviewer Infomation
Scanner data
RFID Scanner

Component-level design

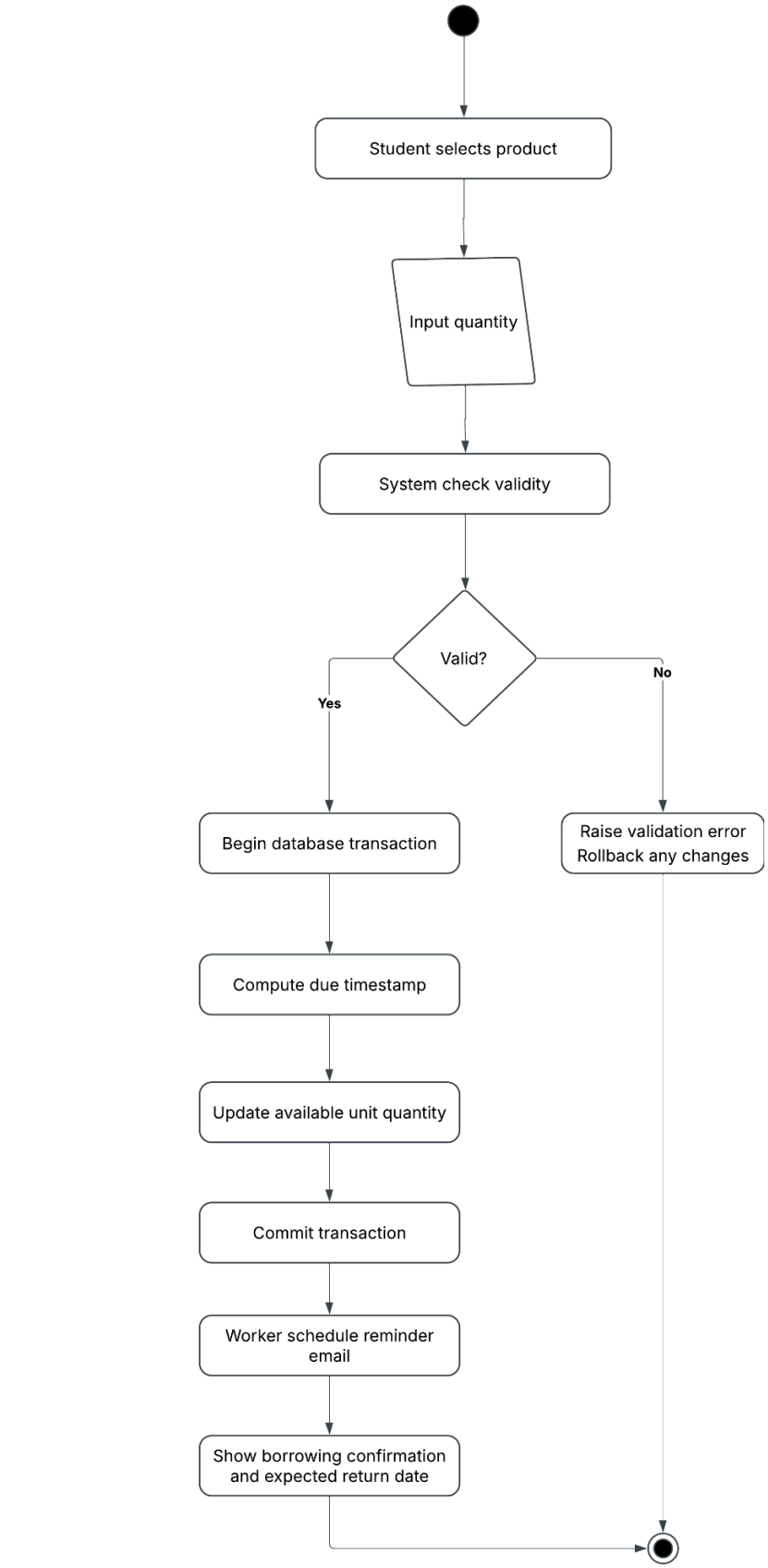
Algorithms



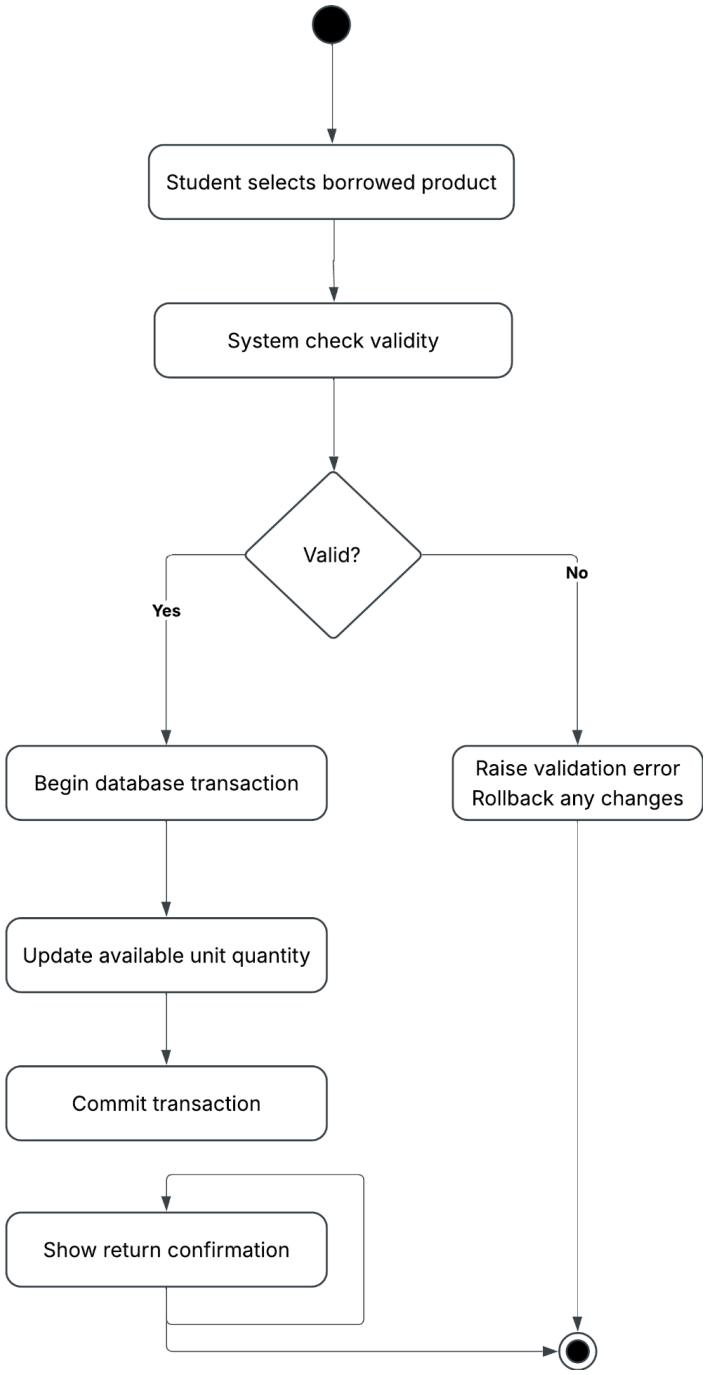
HonestME Store - Stock Update



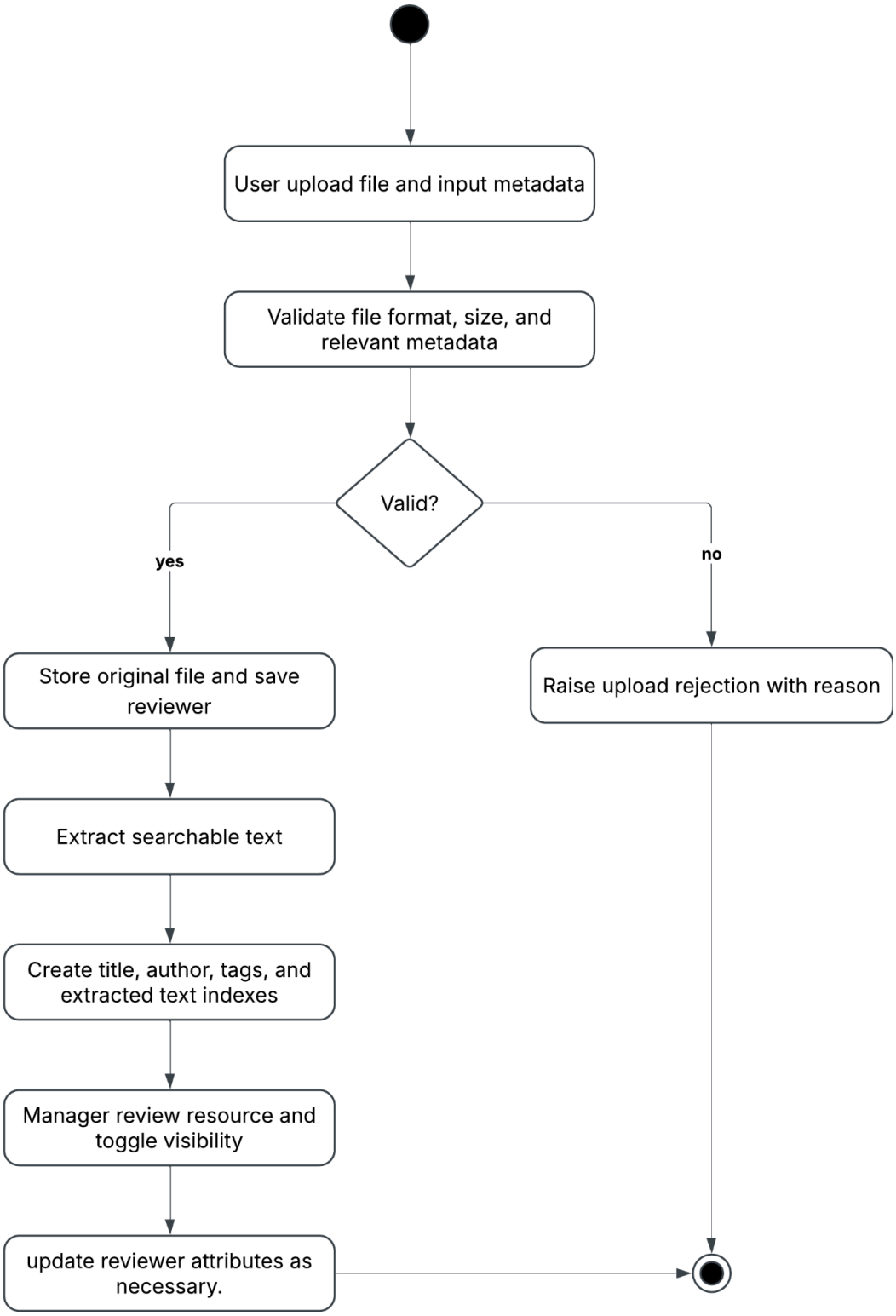
ME-kailangan - Borrowing



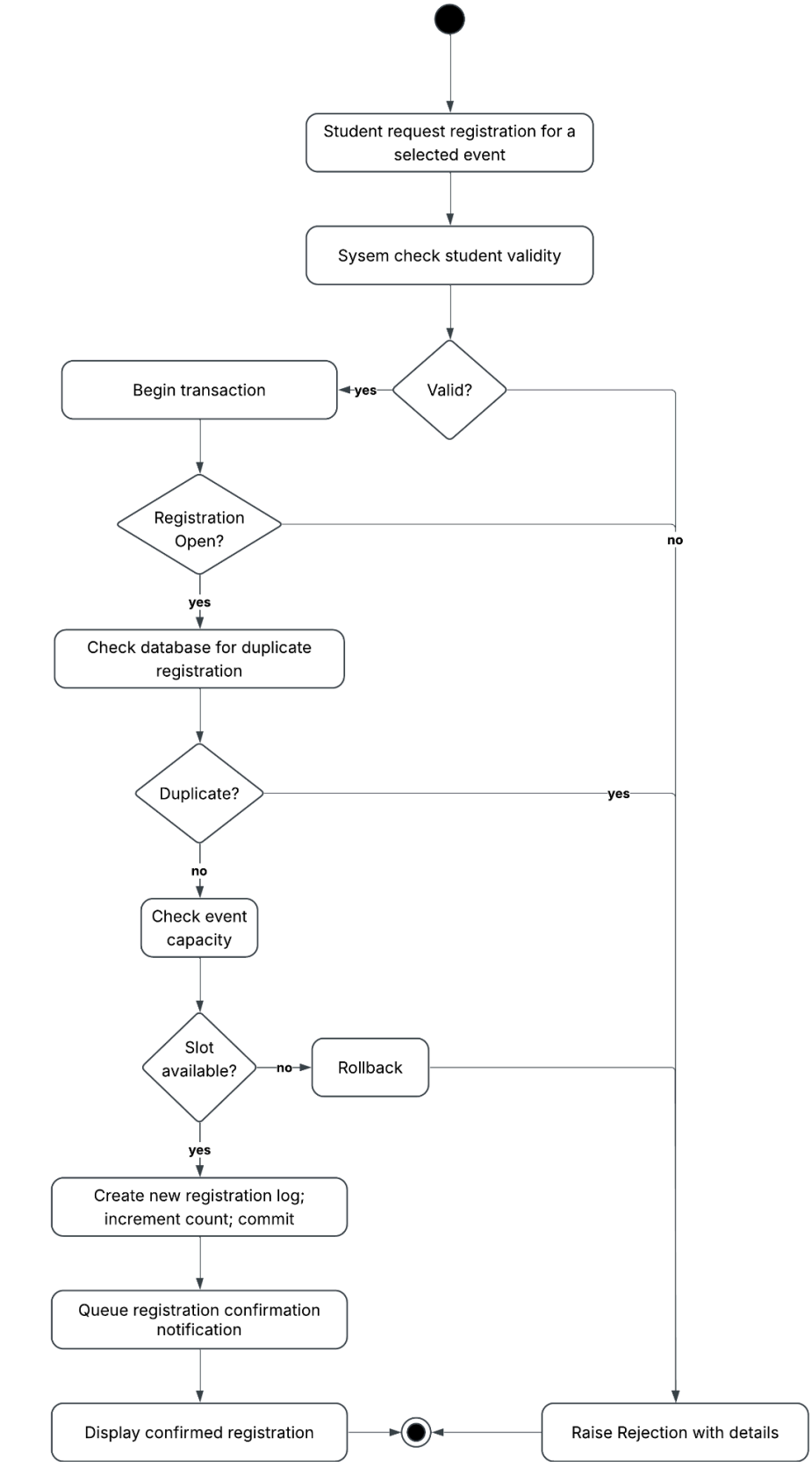
ME-kailangan - Returning



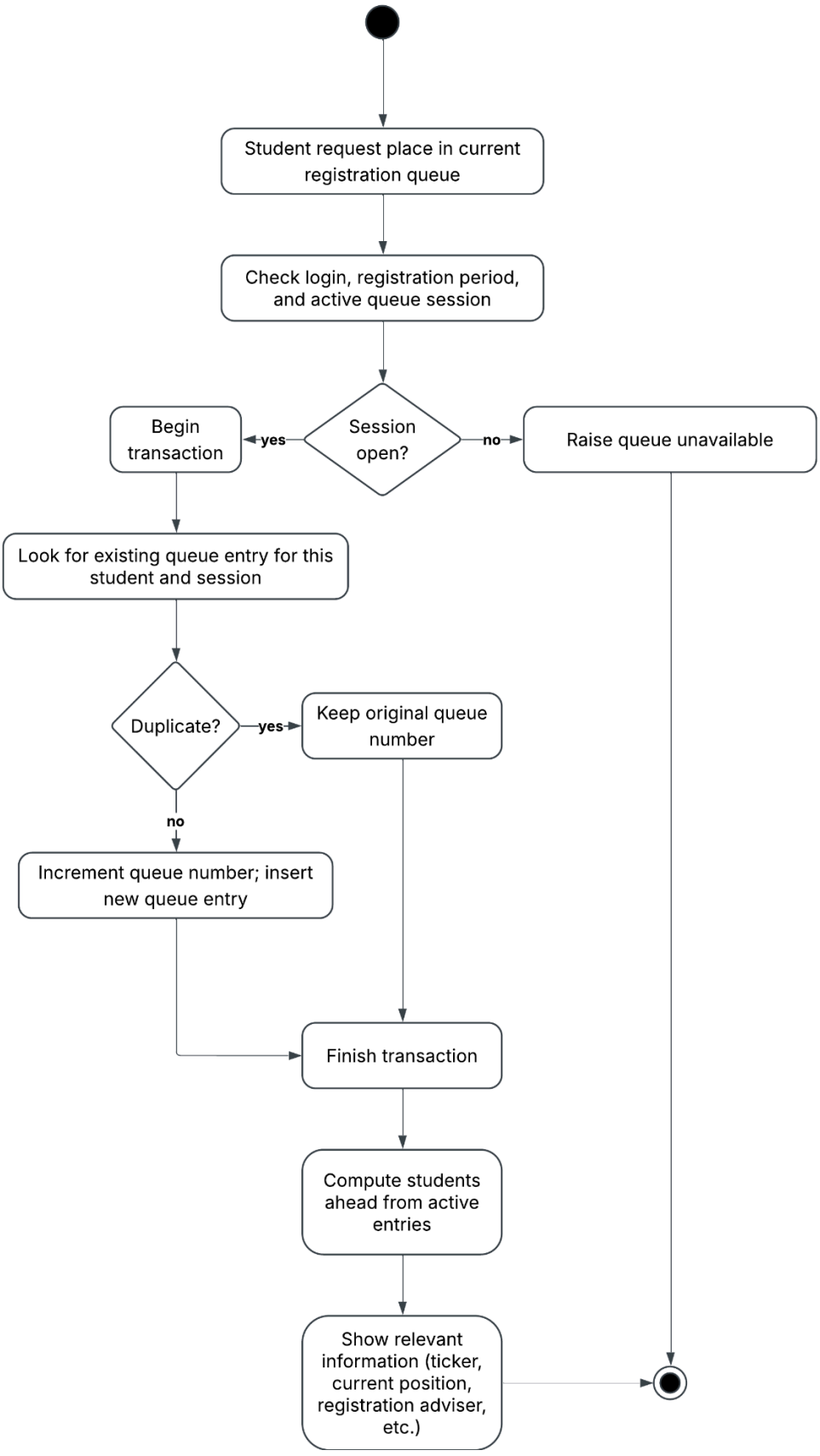
MEg-aral — Reviewer Upload



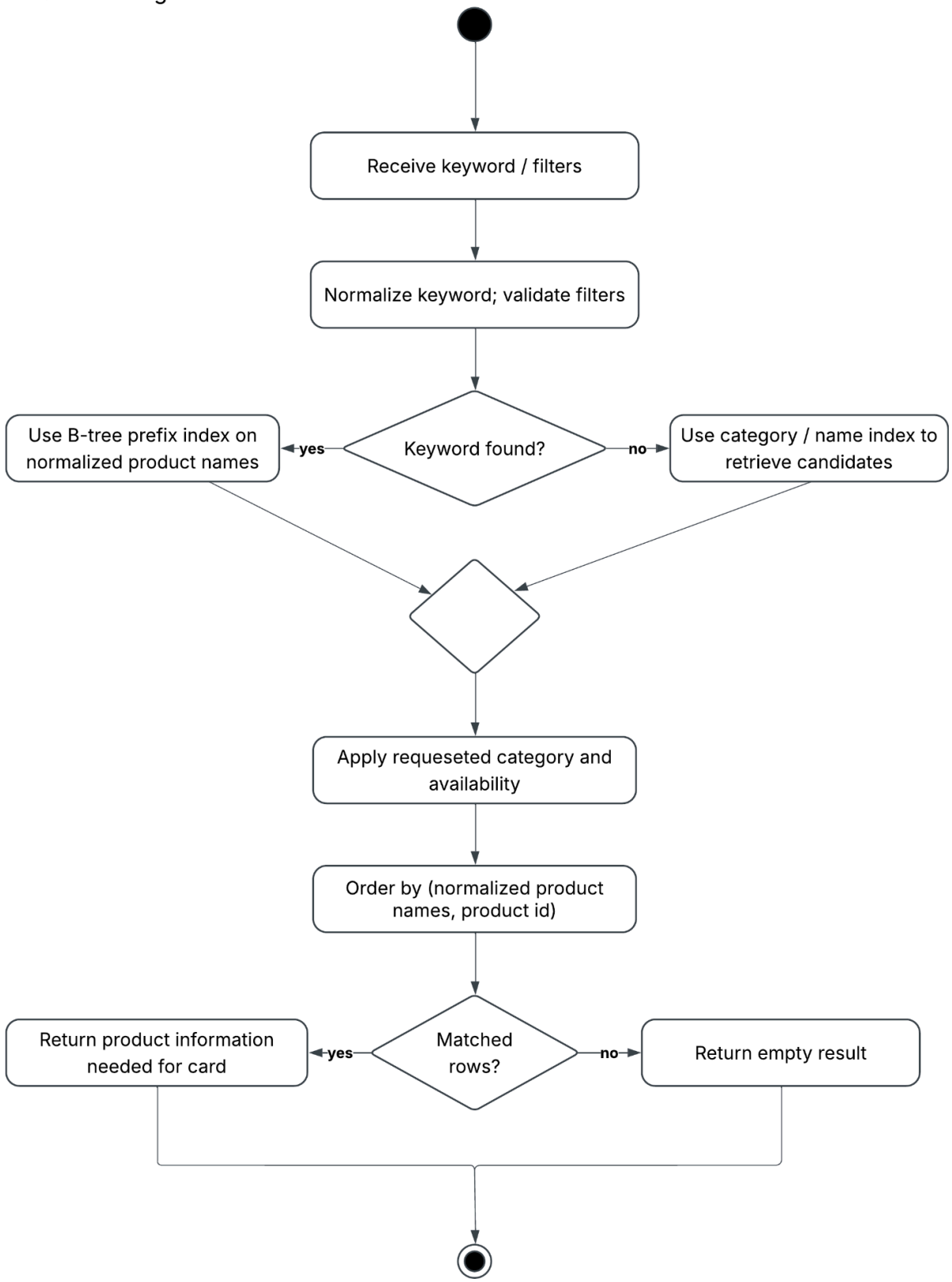
ME Days - Event Registration



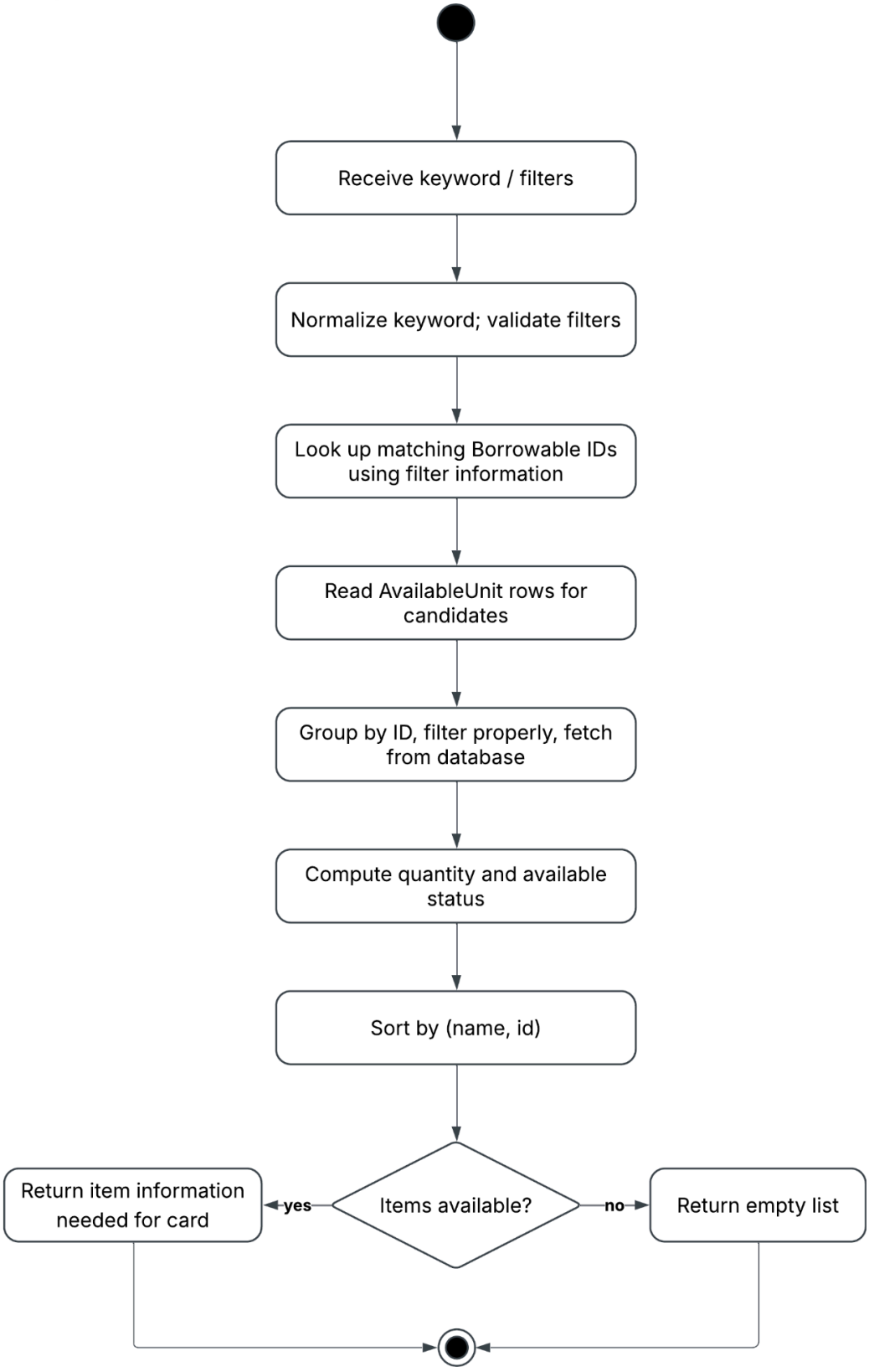
Registration Queue



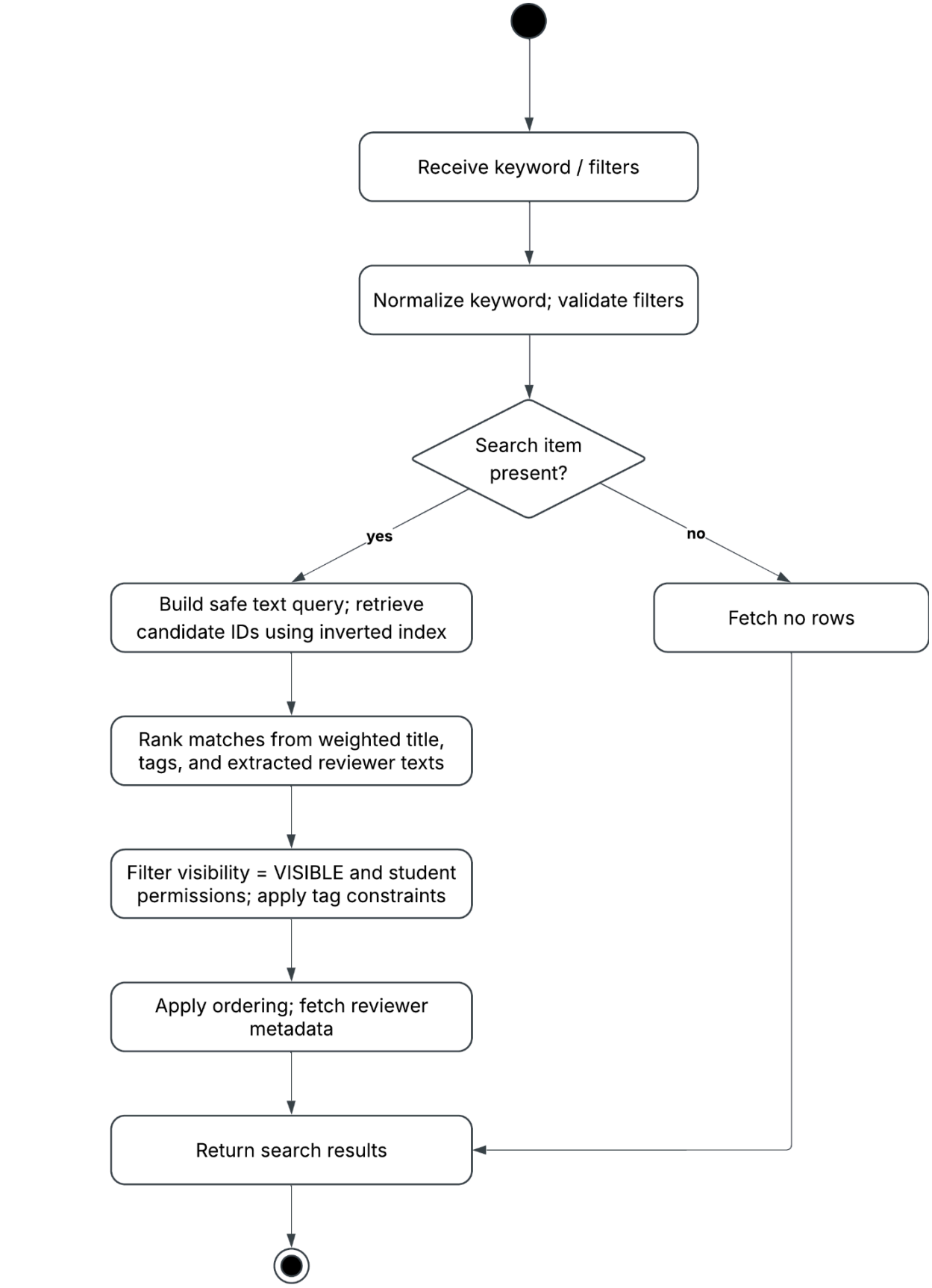
HonestME Store — Indexed product search and filtering



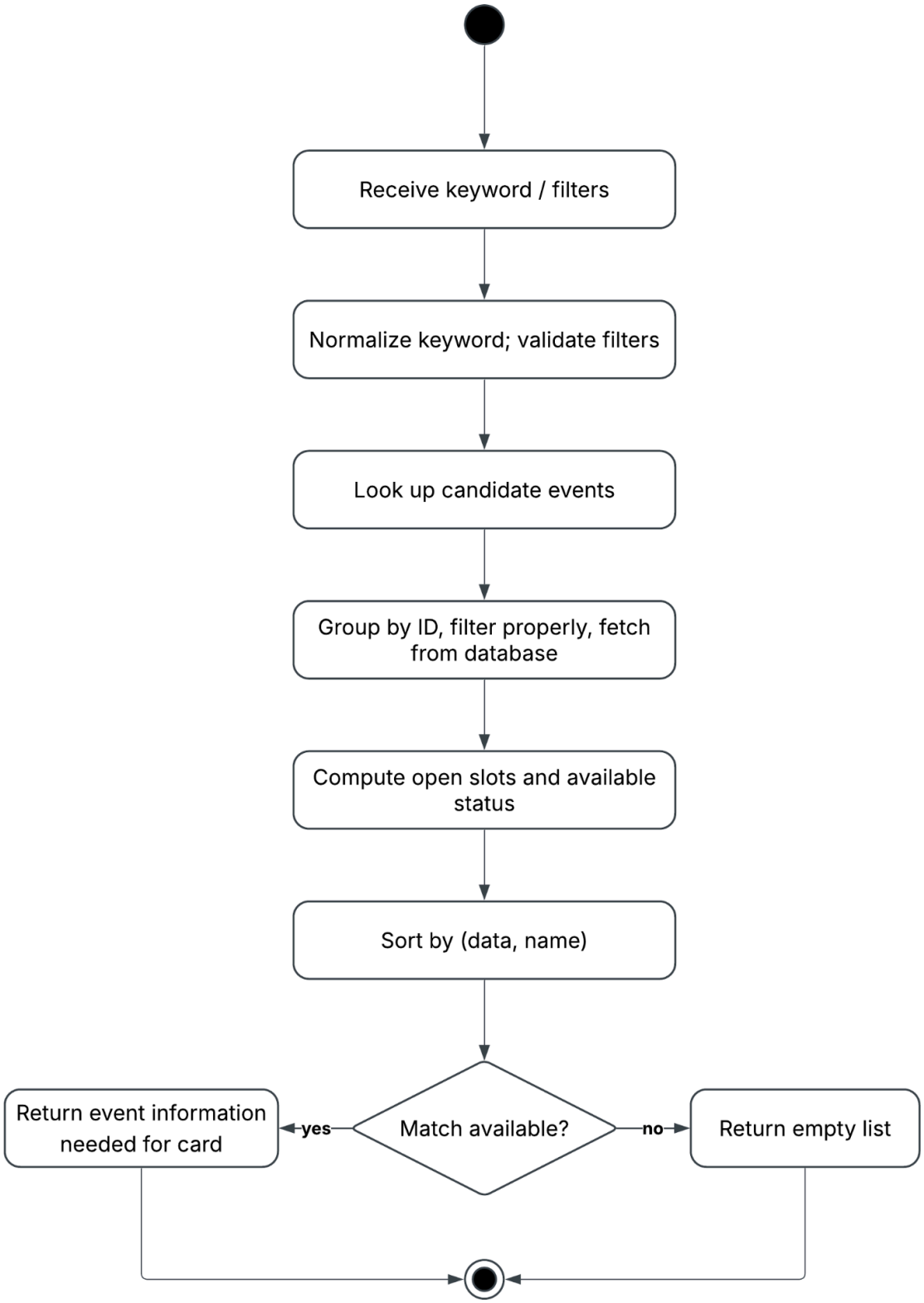
ME-kailangan — Item aggregation
(View, Search, Filter)



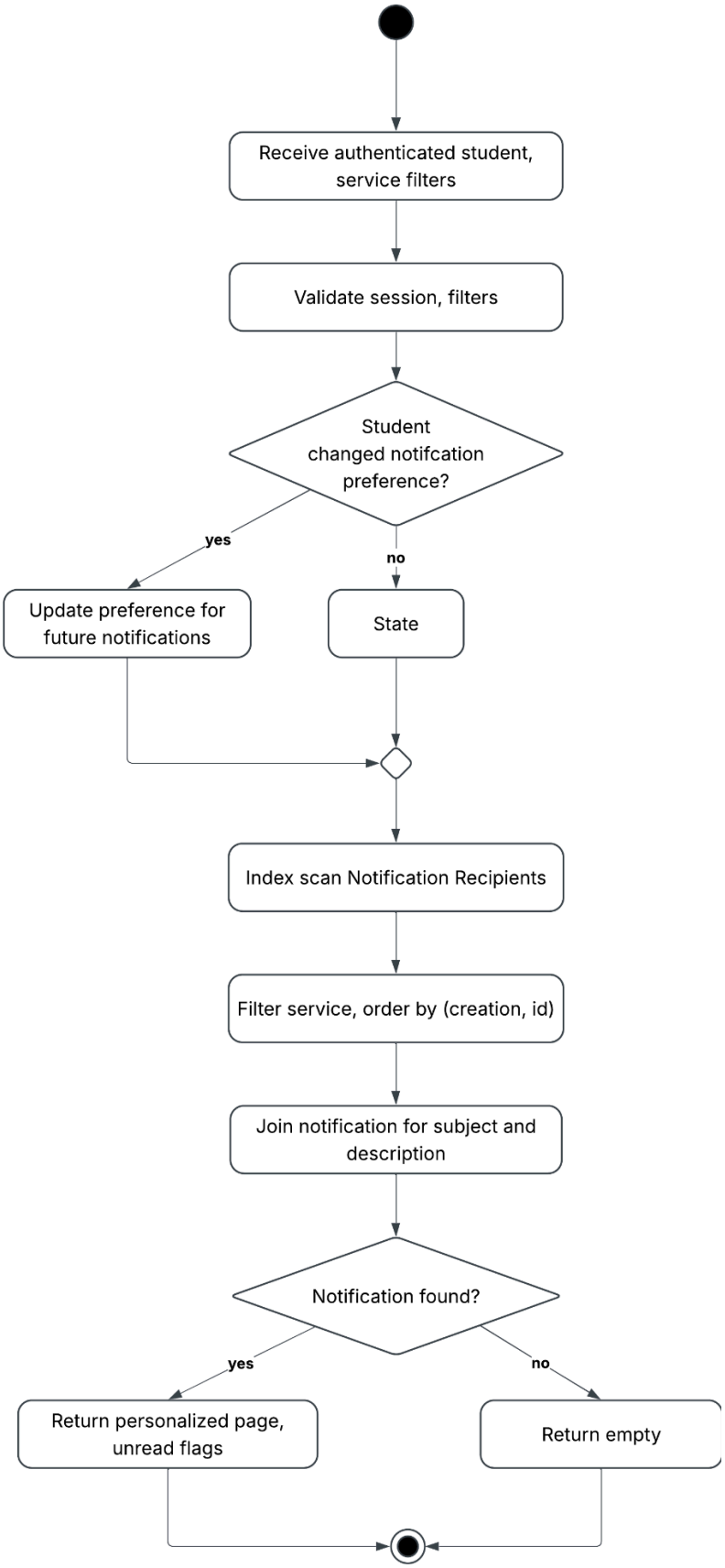
MEg-aral — Item aggregation (View, Search, Filter)



ME-days — event aggregation (View, Search, Filter)



Notification Service



Data Structures

A. Persistent Data/Object Structures

These are data objects that are defined based on the content model, implemented through databases (or local struct definition for storage). This also includes derived attributes if required by the client.

B. Algorithmic Data Structures

Accessing the database repeatedly is a slow process, having data structures that persist inside the program itself helps with faster processing of the required data. We match the operation (lookup, traversal, search, queue, etc.) and choose the appropriate information to access: (1) persistent database record for shared, durable states; (2) in-memory structures for short calculations.

Data Structure	Use	Rationale
Arrays/List	For storing search results, product filtering, etc.	Best for simple iteration and requires ordering/sorting. Works for O(n) traversals.
Hash Maps/Sets	Removing duplications, key-value indexing and mapping, etc.	Used for quick O(1) key lookups.
B-tree (database)	Product names, events, queueing, etc.	For O(log n) index based lookups with result retrieval that support range and sorting.
Inverted Indexing	Search queries	Best for finding documents

Queue (FIFO)	Registration queueing	To maintain a first-come-first-serve rule where needed.
Enums / similar structure	State storing	Restricts invalid transitions
Priority Queue	Notification due processing	Efficient earliest retrieval (although for storage persistence may require database-based implementation instead)

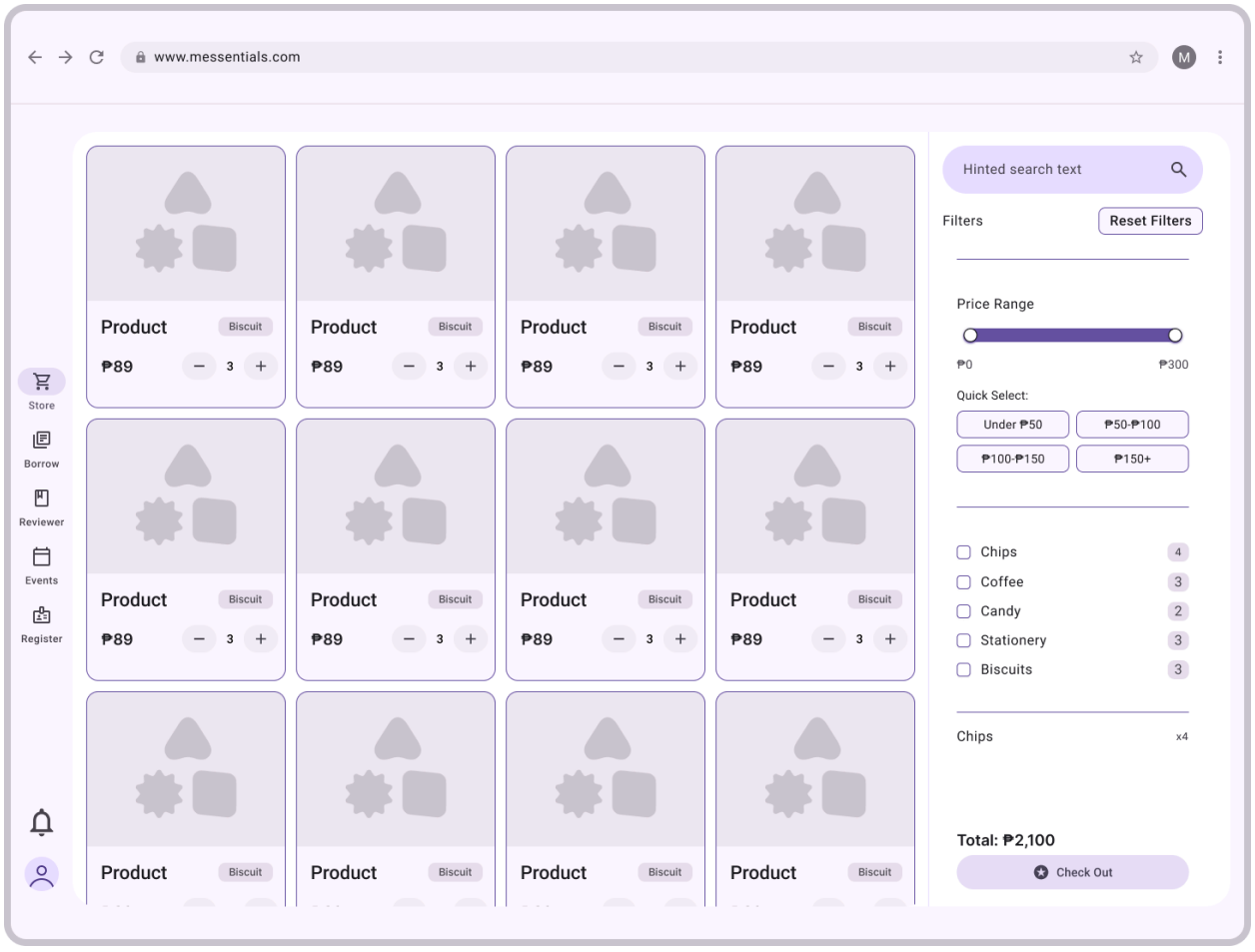
COTS

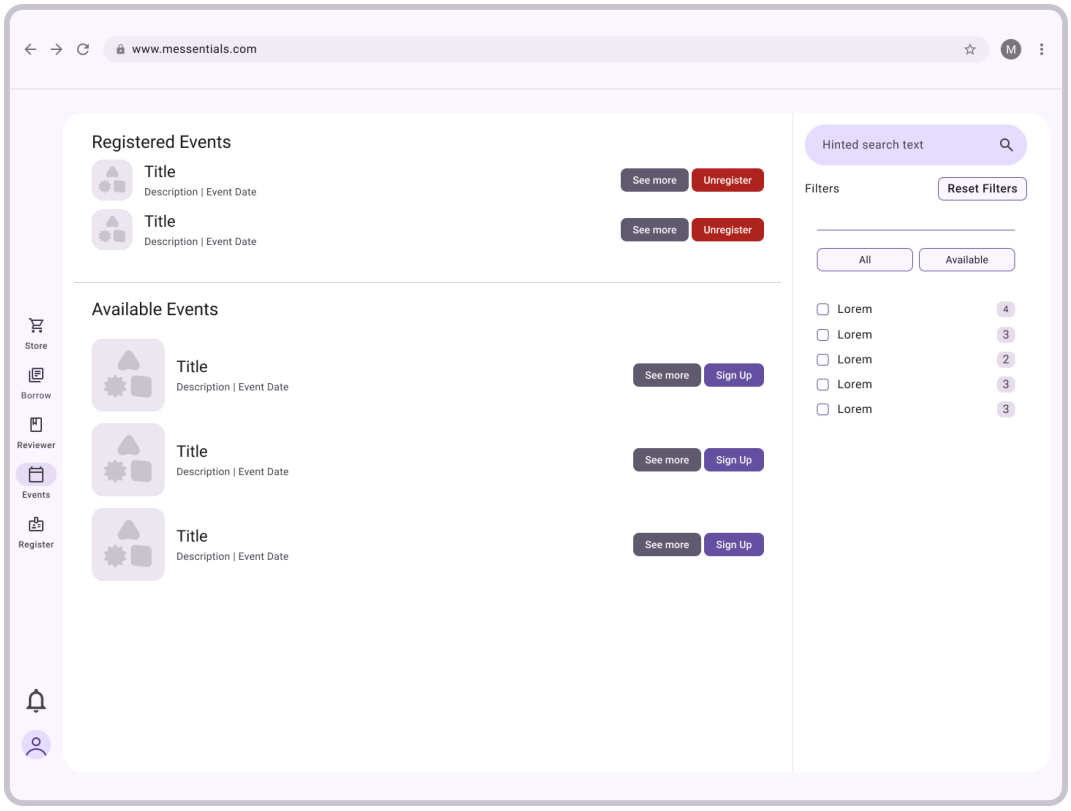
Component	COTS
Identity and authorization	OAuth (e.g. Google) Identity provider
Database updates	Database (PostgreSQL, MySQL, etc.)
Reviewer processing	Database search, Third party file viewers, object storage.
QR Scanning	Third party QR creation
Notification	Background worker (third party if needed), mail services.
FIFO Queue	Database-backed FIFO systems
RFID Scanning	Hardware

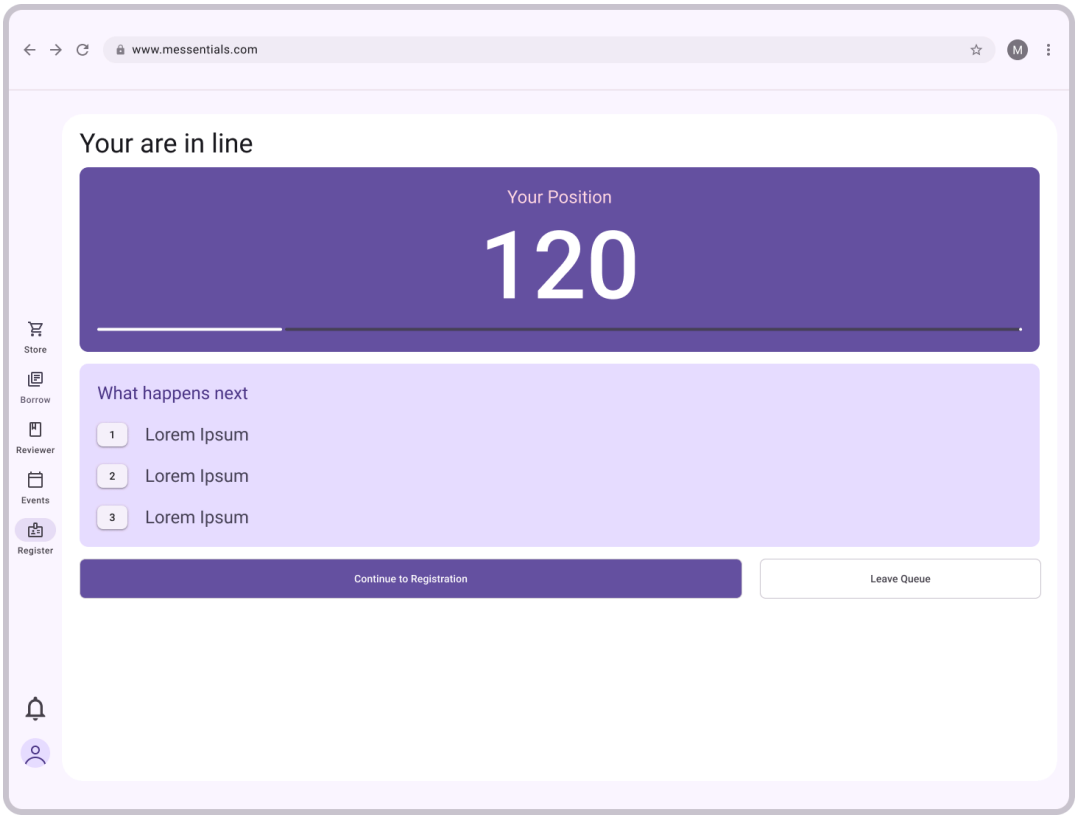
Dependencies

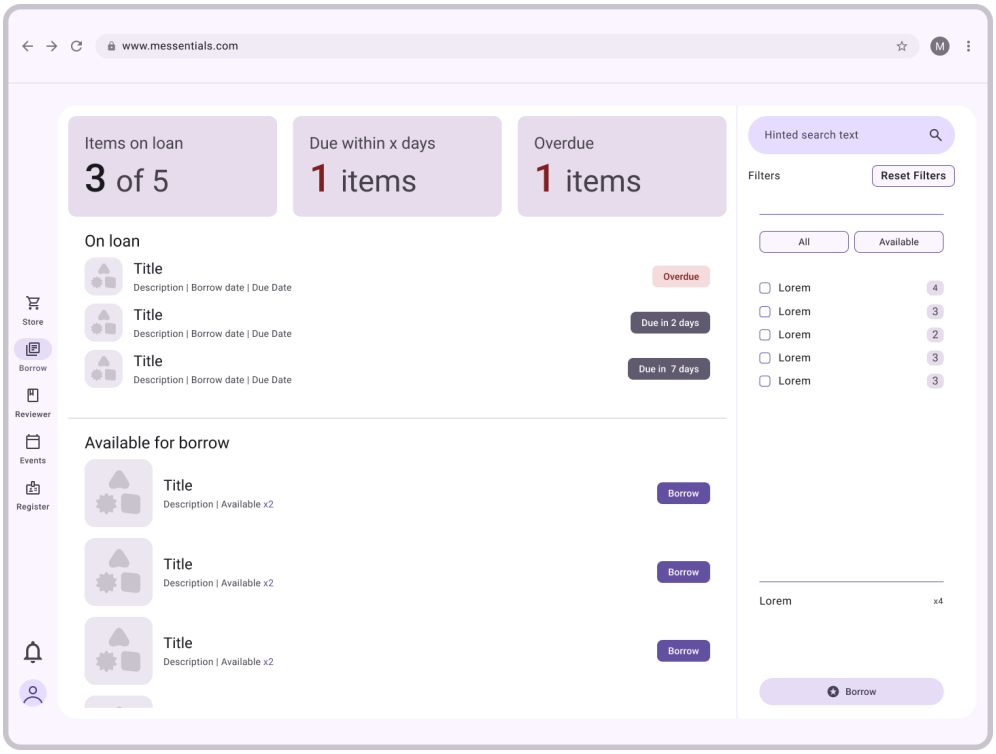
Component	Dependencies
HonestME Store	Authentication, database, notification
ME-kailangan	Authentication, database, notification
MEg-aral	Authentication, file storage, search indexing
ME Days	Authentication, database, notification
Registration Queue	Authentication, database
Access Controls	Authentication, database

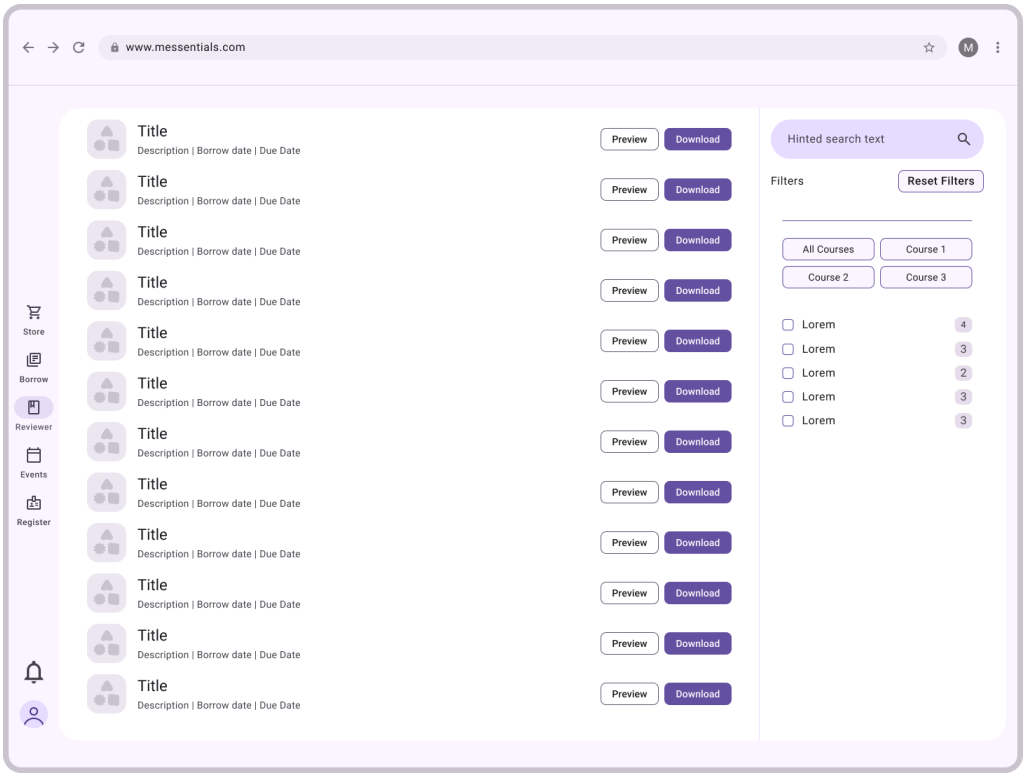
Interface Elements / Usability Design

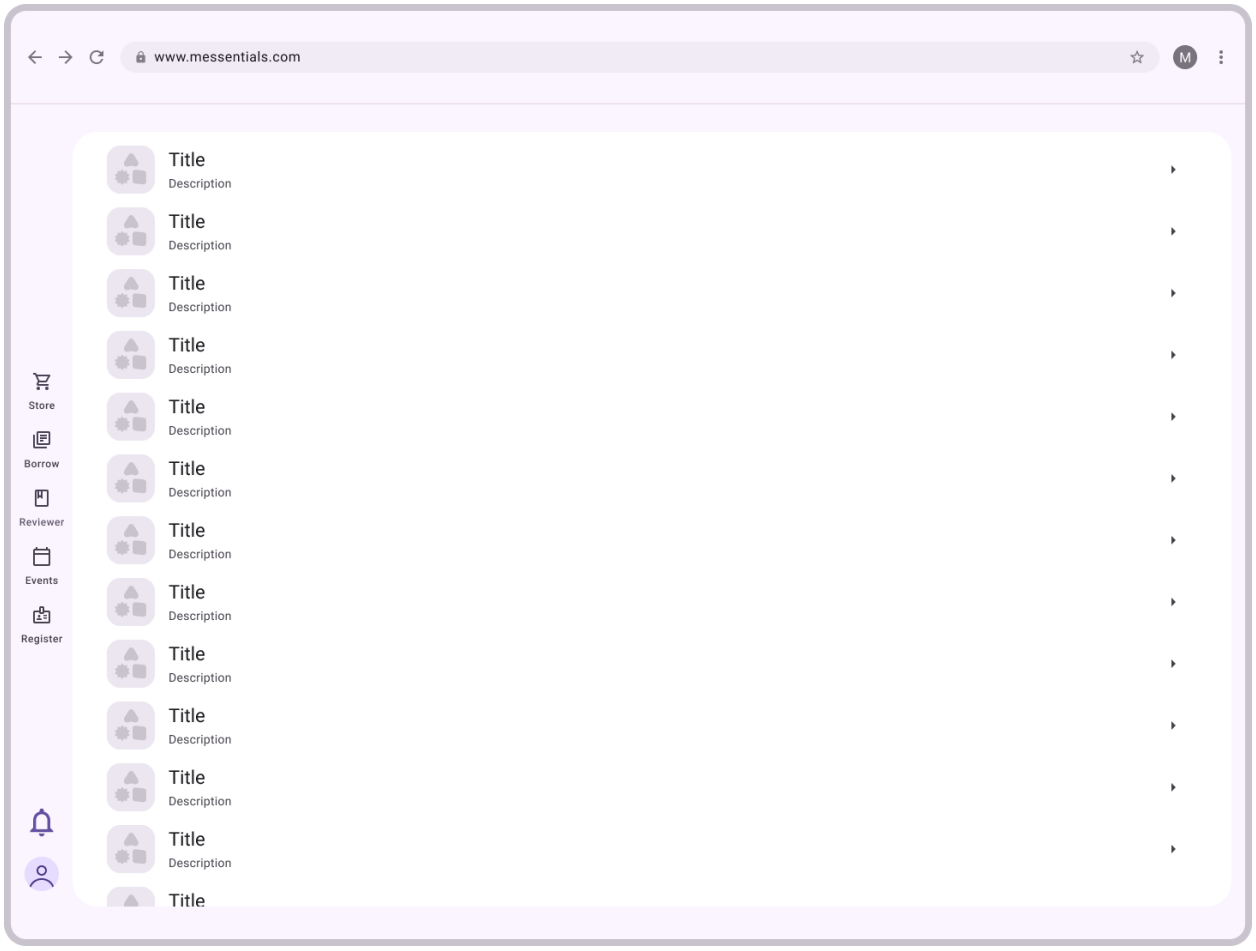



www.messentials.com
Registered Events
Title
Description | Event Date
See more
Unregister
Title
Description | Event Date
See more
Unregister
Available Events
Title
Description | Event Date
See more
Sign Up
Title
Description | Event Date
See more
Sign Up
Title
Description | Event Date
See more
Sign Up
Store
Borrow
Reviewer
Events
Register
Hinted search text
Filters
Reset Filters
All
Available
Lorem 4
Lorem 3
Lorem 2
Lorem 3
Lorem 3


www.messentials.com
Your are in line
Your Position
120
What happens next
1 Lorem Ipsum
2 Lorem Ipsum
3 Lorem Ipsum
Continue to Registration
Leave Queue
Store
Borrow
Reviewer
Events
Register


www.messentials.com
Items on loan
3 of 5
Due within x days
1 items
Overdue
1 items
On loan
Title
Description | Borrow date | Due Date
Overdue
Title
Description | Borrow date | Due Date
Due in 2 days
Title
Description | Borrow date | Due Date
Due in 7 days
Available for borrow
Title
Description | Available x2
Borrow
Title
Description | Available x2
Borrow
Title
Description | Available x2
Borrow
Store
Borrow
Reviewer
Events
Register
Hinted search text
Filters
Reset Filters
All
Available
Lorem 4
Lorem 3
Lorem 2
Lorem 3
Lorem 3
Lorem x4
Borrow


www.messentials.com
Title
Description | Borrow date | Due Date
Preview
Download
Store
Borrow
Reviewer
Events
Register
Hinted search text
Filters
Reset Filters
All Courses
Course 1
Course 2
Course 3
Lorem 4
Lorem 3
Lorem 2
Lorem 3
Lorem 3



Interface Design - Menus and Links

Our interface layout uses a consistent layout based on Material Design 3 components, featuring a left vertical sidebar navigation rail. This shared navigation system is implemented and will provide access to the five main services of this central student hub: HonestME Store, ME-kailangan, MEg-aral, ME Days, and Registration Queueing respectively from top to bottom. The notification and profile controls are positioned on the bottommost of the navigation rail.

Each page features the main content area that occupies the center of the screen and changes accordingly. HonestME Store uses a grid of product cards,

while ME-kailangan, ME Days, MEg-aral, and Notifications primarily use list-based layouts to display their respective records. Registration Queueing uses a status-focused layout that emphasizes the student's queue position and available actions.

Search bars and filter panels are made available immediately, positioned on the right side of applicable services, allowing users to find and narrow down items without leaving the current page. Buttons, cards, and list entries provide all contextual information and navigation needed to their respective item details, processes, call-to-action, and the likes.

The navigation design also follows a consistent structure across the application to reduce the need of relearning the page layout when switching between services.

Aesthetic Design - Layout and Graphics

Our website uses most of its components and graphics from Material Design 3, Google’s open-source design system for building usable products. This provides us a way to have a consistent set of interface components, colors, typography, shapes, and layout guidelines across our web applications. With the help of this Design guide, we aim to provide the users a clean and user-friendly interface that are tested by professionals that helps them navigate through the website with ease.

The focus of the component layout for our website is ease of use, giving a compact feeling for students helping them see as much as possible without

having to move around the website to gather new information. This is evident in the sample designs given where the search and filter functions are directly open on the right side of the layout, all the while in the sample for most of the page that we show as many products as possible.

Layout

The web application will also be updated to follow a responsive layout to adapt with different screen sizes, with the starting template being available on desktop only for now just to get a general feel of the website before committing to other adjustments.

Following the guidelines in Section 13.5, related elements will be grouped together, and sufficient whitespace will be maintained to avoid clutter. Important information and frequently used actions will be placed in prominent areas, while less frequently accessed details will be organized accordingly. To reduce excessive scrolling we also made the design as compact as possible.

Graphics

The application will use the Material Design 3 color system, which already has predefined primary, secondary, and tertiary color roles. Primary colors are used to emphasize important actions, guiding the user to its call-to-action, while surface colors are used to distinguish between the available contents and components. Proper colors will also be used to indicate states of the components which are not limited to: hovering, unavailable, clicked, pressed, etc.

The typography also follows Material Design 3's scale, this includes the display, headlines, titles, body, and label styles. These predefined values help in differentiating the contents with the help of different font sizes, weight, and spacing to establish clear visual hierarchy.

The components used are also from Material Design 3 which includes button designs, labels, cards, search bars, menus, and elements / icons. This makes sure the shapes stay consistent as well as the values for their padding, rounding, elevation, etc. to establish uniformity in the appearance throughout the website.

The website will also maintain adequate contrasts between the colors for better visual feedback, with proper description labels. Proper communication is also established with the help of proper icons for the use (e.g. shopping cart for the store) and the proper usage of colors (e.g. using red for errors).

Overall, our aesthetic design prioritizes clarity, consistency, accessibility, and efficient use of screen space, in accordance with the layout and graphic design principles discussed in Section 13.5. Using Material Design 3 provides us with a unified visual foundation while allowing each student service to use the interface elements most appropriate to its functions.

Navigation Design

Within each service, students can navigate through search bars, filtering controls, selectable items, and action buttons. The layout of the navigation prioritizes the commonly used functions, allowing students to access services directly without the need to navigate through other parts like the home page, a new pop-up for filters, etc. Additionally, the filtering and search system helps the students by allowing them to locate specific content without needing to go through the whole catalog.

The website follows with accordance to Section 7.5.8, specifically: Certain elements are easier to reach depending on priority, emphasis on call to actions, errors such as invalid searches and unavailable actions are handled through clear error messages/disabled components, related groups of elements are consistent, navigation are a combination of different navigational means, and the likes.

Navigation Scenario Analysis

Consider the following assumptions:

- The student is already logged in and begins on a service page (WLOG)
- Each mouse click is an interaction and a character type is one keystroke
- Search queries count as navigation input
- Quantity adjustments are clerical data entry
- Information not provided in the earlier wireframes are not counted.

Scenario	Mouse clicks	Keystrokes	Navigation	Data entry
HonestME Store: Search for a product, adjust quantity, and select Check Out	5	6	3	2
ME-kailangan: Filter available items, select an item, and initiate borrowing	4	0	4	0
ME Days: Filter available events and select Sign Up	3	0	3	0
Registration Queueing: View queue status and select Leave Queue	2	0	2	0
MEg-aral: Search for a reviewer and select Preview	3	4	7	0
Notifications: Open the notification page and select a notification	2	0	2	0
Total	19	10	21	2

Scenarios

1. HonestME Store: click the navigation rail icon for store (1 click), click on the search bar (1 click), enter item name (average ~6 letters), increase selected quantity (on average, a person buys at least 2 items from the honesty store, so 2 clicks), select check out (1 click).
2. ME-kailangan**:** open the Borrow page (1 click), select the Available filter (1 click), choose the Borrow button beside an item (1 click), and select the

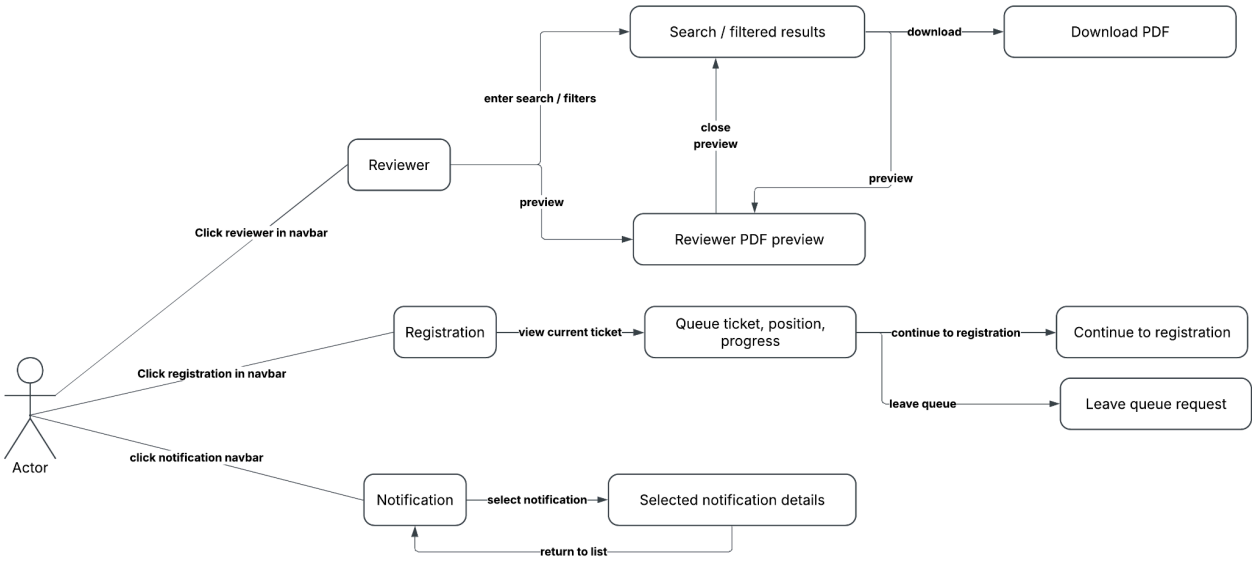
Borrow button at the bottom of the interface (1 click). This requires 4 interactions.

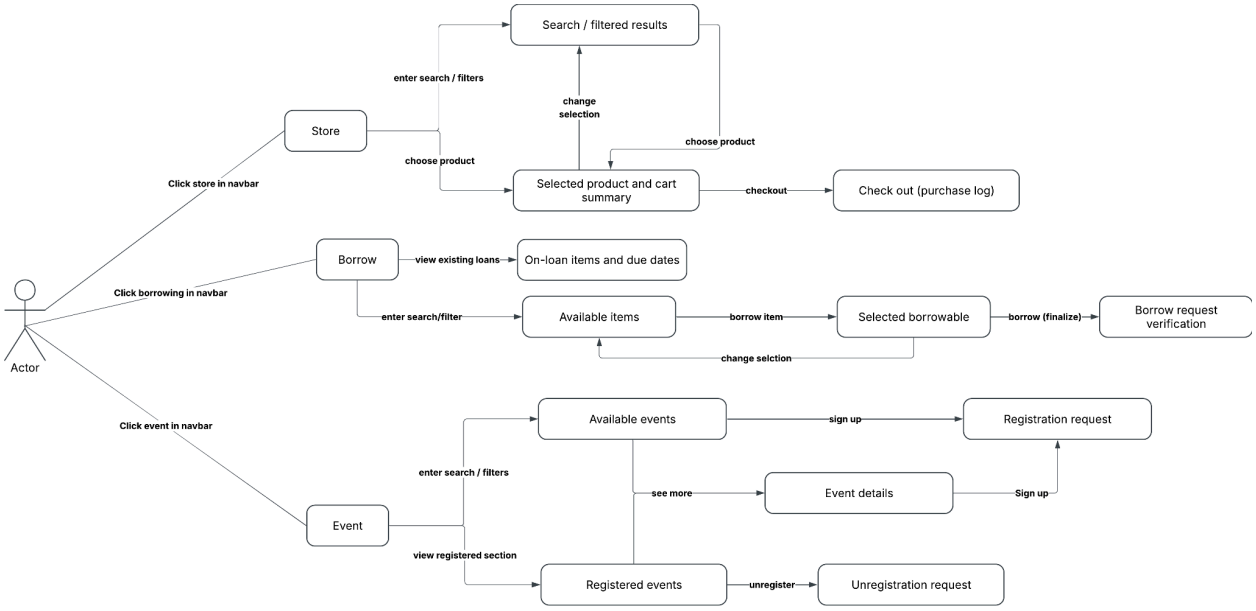
3. ME Days: open Events (1 click), apply the Available filter (1 click), and select Sign Up for an available event (1 click). This requires 3 interactions.
4. Registration Queueing: assume the student is already in the queue and opens the Register page (1 click). The queue position is displayed without further input. Selecting Leave Queue requires one additional click, resulting in 2 interactions.
5. MEg-aral: open Reviewer (1 click), focus the search bar (1 click), enter "Math" (4 keystrokes), and select Preview (1 click). This requires 7 interactions.
6. Notifications: select the notification icon (1 click) and open an individual notification (1 click). This requires 2 interactions.

Navigation Semantic Units

NSU	User Goal	Navigation Pathway
1: HonestME Store	Locate products and log purchases	Store Catalog → Search/Filter → Product Selection → Cart Summary → Check Out
2: ME-kailangan	Locate and borrow an item	Borrowing Page → Available Items → Item Selection → Borrow Request
3: ME Days	Discover events and register	Event Listing → Available Events → Event Details or Sign Up → Registration Request
4: Registration Queue	Track registration position	Registration Page → Queue Ticket → Continue to Registration or Leave Queue
5: MEg-aral	Locate academic resources	Reviewer Library → Search/Filter → Reviewer Preview or Download
6: Notifications	Read received notifications	Notification List → Selected Notification Details

NSU Diagrams




Search / filtered results
enter search / filters
change selection
Store
choose product
choose product
Click store in navbar
Selected product and cart summary
checkout
Check out (purchase log)
Borrow
view existing loans
On-loan items and due dates
Click borrowing in navbar
enter search/filter
Available items
borrow item
Selected borrowable
borrow (finalize)
Borrow request verification
change selction
Actor
Click event in navbar
Available events
sign up
Registration request
enter search / filters
see more
Event details
Sign up
Event
view registered section
Registered events
unregister
Unregistration request